Supplement: Computational-Graph Interpretation of NNS and ViT Structural Paths

Return to manuscript. This file contains reproducibility parameters, complete statistics, archived figure navigation, and prospective experiment scope. RQ labels in the current manuscript follow the revision.

S1. Model and registry details

Models: `vit_base_patch16_224.augreg2_in21k_ft_in1k` and `deit_small_patch16_224.fb_in1k`. Torch 2.2.2+cu121; timm 0.9.16; torchvision 0.17.2+cu121; NumPy 1.26.4; Pillow 10.2.0.

Table S1. Original scalar registry coverage.

Model	Object	Candidate definition	Candidates	Registered/image	Coverage	Strategy	Scalar neuron
vit_b16	post-block token-channel	12*197*D	1,815,552	192	0.01058%	12 blocks x 4 tokens x 4 channels	yes
vit_b16	post-block patch sum/energy	2*12*D	18,432	96	0.52083%	12 blocks x 4 channels x 2 aggregates	no
vit_b16	attention-head coordinate	12*H*197*64	1,815,552	144	0.00793%	all heads; CLS token; within-head channel 0	yes
vit_b16	module patch aggregate	3*(4D+24*D)	27,648	18	0.06510%	blocks 0/5/11; six modules; channel 0	no
vit_b16	patch embedding aggregate	D	768	4	0.52083%	four fixed channels; sum over patches	no
vit_b16	final CLS coordinate	D	768	1	0.13021%	channel 0	yes
vit_b16	ground-truth logit	1000	1,000	1	0.10000%	frozen image label	output
vit_b16	clean-top1 logit	1000	1,000	1	0.10000%	frozen clean prediction	output
deit_s16	post-block token-channel	12*197*D	907,776	192	0.02115%	12 blocks x 4 tokens x 4 channels	yes
deit_s16	post-block patch sum/energy	2*12*D	9,216	96	1.04167%	12 blocks x 4 channels x 2 aggregates	no
deit_s16	attention-head coordinate	12*H*197*64	907,776	72	0.00793%	all heads; CLS token; within-head channel 0	yes

Model	Object	Candidate definition	Candidates	Registered/image	Coverage	Strategy	Scalar neuron
deit_s16	module patch aggregate	3(4D+24D)	13,824	18	0.13021%	blocks 0/5/11; six modules; channel 0	no
deit_s16	patch embedding aggregate	D	384	4	1.04167%	four fixed channels; sum over patches	no
deit_s16	final CLS coordinate	D	384	1	0.26042%	channel 0	yes
deit_s16	ground-truth logit	1000	1,000	1	0.10000%	frozen image label	output
deit_s16	clean-top1 logit	1000	1,000	1	0.10000%	frozen clean prediction	output

Table S2. Resources and extended validation coverage.

Resource / validation	Value
Formal scan wall time	1,752.71 s
Four-card GPU-hours upper bound	1.9475
Throughput	192.16 targets/s
Retained data	9.290 GB
Peak reserved GPU memory	2.284 GB
Peak CPU RSS	2.018 GB
M1 random permutations	128,000,000
M4 attention / residual / JVP records	86,400 / 19,200 / 691,200

S1.1 Implementation and core-metric pseudocode

All models run in evaluation mode. M3 computes chunked VJPs with torch.autograd.grad in is_grads_batched mode and frozen gradient_chunk_size=16. Evidence JSON files record random seeds, model and image hashes, worker environments, and per-task times. Table S2 reports scan time, throughput, memory, and retained storage; the 456 supplemental-control tasks total 2.4502 GPU-hours. Core implementations reside in src/vitnns/endpoint_m3_atlas.py, src/vitnns/endpoint_m4.py, and the corresponding scripts/run_endpoint_*.py entry points.

```
Endpoint(s, x):
    g <- VJP(s, x) at the observed forward state
    return x * g

StructuredCut(F, edges, reconstruction_target):
    for each actual use-edge z_k in a valid cut:
        g_k <- downstream cotangent dF/dz_k
        E_k <- x * VJP(z_k, x; vector=g_k)
    assert sum_k E_k closes to reconstruction_target # branch contribution or Total Endpoint

SpatialMetrics(E, target_patch):
    M[p] <- sum_rgb abs(E[p, rgb])
    if sum_p M[p] == 0: return NA
    p_patch <- patch_sum(M) / sum_p M[p]
```

```
    entropy <- -sum_j p_patch[j] log(p_patch[j]) / log(196)
    locality <- neighborhood_mass(target_patch) - uniform_neighborhood_mass
    distance <- sum_j p_patch[j] normalized_distance(j, target_patch)

RandomRegionAUC(repeats):
    for each random repeat r:
        curve_r <- absolute finite effect at every area budget
        auc_r <- trapezoid(curve_r, area in [0, 0.25]) / 0.25
    return mean_r auc_r
```

CLS has no natural target patch, so locality and target-relative distance are defined only for patch-token targets; zero-mass cases propagate as NA. Tokens, channels, and heads are aggregated within each image by prespecified rules before image-level inference.

S2. Frozen decisions and numerical diagnostics

C4a: BROAD_BUT_NON_UNIVERSAL_FAMILY_LEVEL_SUPPORT; original all-family omnibus NOT_SUPPORTED.
C4b: NOT_SUPPORTED. The numerical atlas status is PASS_FLOAT64_RECOVERY_WITH_RETAINED_FLOAT32_FAI
36/57,600 float32 failures are retained and 36/36 same-coordinate float64 checks pass. These fields are not interchangeable.

Table S2.1. Historical pooled C4a provenance; image-level median [95% CI]. Holm p values belong to the original joint family of 28 component and regional tests.

Model	Reference	Heads	A/V pooled	Residual pooled
vit_b16	normalized-zero	0.291 [0.281, 0.298]	0.0996 [0.075, 0.123]	0.177 [0.155, 0.195]
vit_b16	blur	0.306 [0.296, 0.318]	0.157 [0.131, 0.177]	0.209 [0.182, 0.237]
deit_s16	normalized-zero	0.191 [0.177, 0.205]	0.107 [0.078, 0.129]	0.135 [0.113, 0.164]
deit_s16	blur	0.182 [0.168, 0.195]	0.160 [0.128, 0.185]	0.229 [0.204, 0.251]

All 12 original Holm-adjusted p values are below 10^{-6}. Eleven of twelve families met the original composite operational threshold, while all-family C4a remained `NOT_SUPPORTED`. This table preserves the historical decision identity; the S18 stratification does not revise it retrospectively.

S3. Reference sensitivity

Change uses atol=1e-12 and rtol=0. Relative change is abs(q0-qblur)/max(abs(q0),abs(qblur)); both-zero denominators are NA. Component quantiles describe the pooled records, not independent inferential samples.

Table S3. Changes in Taylor magnitude and absolute finite effect.

Model	Quantity	Component pairs	Above tolerance	Abs change median [q10,q90]	Relative change median [q10,q90]
DeiT-S/16	finite_effect	67200	100.0%	0.1365 [0.0161, 1.6575]	0.4927 [0.0985, 0.8997]
DeiT-S/16	taylor	67200	100.0%	0.1215 [0.0150, 0.7867]	0.5186 [0.1172, 0.9033]
ViT-B/16	finite_effect	96000	100.0%	0.0713 [0.0076, 1.4414]	0.5300 [0.1117, 0.9137]

Model	Quantity	Component pairs	Above tolerance	Abs change median [q10,q90]	Relative change median [q10,q90]
ViT-B/16	taylor	96000	100.0%	0.0688 [0.0078, 0.5471]	0.5225 [0.1133, 0.9052]

Head rankings use average ranks and Kendall tau-b. Top-K ties are broken by ascending component ID; Top-3 overlap is intersection size / 3. Rank displacement uses original rank positions, so model head counts affect its scale. Each image averages 12 blocks, then summaries use 400 images. Full-order equality counts are Taylor 0/4,800 and effect 0/4,800 for ViT-B, Taylor 46/4,800 and effect 64/4,800 for DeiT-S.

Table S4. Image-level rank concordance distributions.

Model	Quantity	Metric	Image median [q10,q90]	n / NA
DeiT-S/16	finite_effect	spearman	0.3071 [0.1238, 0.4524]	400 / 0
DeiT-S/16	finite_effect	kendall_tau_b	0.2444 [0.0989, 0.3678]	400 / 0
DeiT-S/16	finite_effect	top1_agreement	0.4167 [0.1667, 0.5833]	400 / 0
DeiT-S/16	finite_effect	top3_overlap	0.6111 [0.5278, 0.6944]	400 / 0
DeiT-S/16	finite_effect	mean_absolute_rank_displacement	1.5278 [1.2778, 1.7528]	400 / 0
DeiT-S/16	taylor	spearman	0.3000 [0.1000, 0.4476]	400 / 0
DeiT-S/16	taylor	kendall_tau_b	0.2389 [0.0778, 0.3667]	400 / 0
DeiT-S/16	taylor	top1_agreement	0.3333 [0.1667, 0.5833]	400 / 0
DeiT-S/16	taylor	top3_overlap	0.6111 [0.5278, 0.6944]	400 / 0
DeiT-S/16	taylor	mean_absolute_rank_displacement	1.5278 [1.3056, 1.8056]	400 / 0
ViT-B/16	finite_effect	spearman	0.3759 [0.2622, 0.4904]	400 / 0
ViT-B/16	finite_effect	kendall_tau_b	0.2803 [0.1967, 0.3740]	400 / 0
ViT-B/16	finite_effect	top1_agreement	0.3333 [0.2500, 0.5833]	400 / 0
ViT-B/16	finite_effect	top3_overlap	0.5000 [0.4167, 0.6111]	400 / 0
ViT-B/16	finite_effect	mean_absolute_rank_displacement	2.9444 [2.6111, 3.2917]	400 / 0
ViT-B/16	taylor	spearman	0.3782 [0.2539, 0.4914]	400 / 0
ViT-B/16	taylor	kendall_tau_b	0.2828 [0.1942, 0.3737]	400 / 0
ViT-B/16	taylor	top1_agreement	0.3333 [0.2500, 0.5833]	400 / 0
ViT-B/16	taylor	top3_overlap	0.5000 [0.4167, 0.6111]	400 / 0
ViT-B/16	taylor	mean_absolute_rank_displacement	2.9444 [2.5972, 3.2792]	400 / 0

S4. Method objects

Table S5. Functional comparison.

Method	Primary object	External reference	Propagation / contract
Endpoint	Normalized-input Total Endpoint	No; normalization origin matters	Global cuts close to total; branch partitions close to branch
Activation-space Taylor	Specified replacement effect	Yes	Clean cotangent × displacement
Integrated Gradients	Baseline-to-input attribution	Yes	Path-average gradient; baseline-difference completeness
Attention rollout	Token flow	No	Accumulated attention / residual flow
Grad-CAM	Class-conditioned coarse localization	No	Gradient-weighted feature maps
Chefer / AttnLRP	Transformer relevance	Implementation-specific	Method-specific relevance rules
Raw attention	Token allocation	No	Forward attention weights

Table S5a. Complete nearest-method matrix. "Not reported" is limited to the versions listed in the references; closure objects are not interchangeable across methods.

Method family	Explained object	Consumer edge/use	Spatial component map	Closure object	Finite intervention
Gradient × Input	Current-point input-weighted gradient	Not reported	One input map	Generally no function-value completeness	No
Integrated Gradients	Reference-to-input path attribution	Not reported	One input map	Output difference	No
Conductance	Path attribution through hidden units	Hidden-unit set	Optional input resolution	Layerwise output difference	No
GlobEnc / ALTI	Cross-layer token contribution/mixing	Operator-level	Token map	Local forward decomposition	No
DecompX	Token components of representations and logits	Component-level	Token components	Recomposition under a bias rule	No
AttnLRP / HiLRP / GradSkip	Transformer relevance	Operator links	Relevance map	Relevance conservation	No
EAP / AtP* / Vi-CD	Reference-specific component or edge effect	Yes	Usually not the input-coordinate components used here	Patching effect or circuit fidelity	Yes
Activation patching / CAAP	Internal-activation replacement effect	Node or specified activation	Patch score	Generally nonadditive	Yes
DAVE	ViT input-gradient decomposition	No consumer-use index reported	Pixel map	Effective input-output map	No
Unpack	Component interactions, named paths, and token credit	Component/path identity	Token attribution	Method-specific credit decomposition	No
NNS–ViT	Same current-point local differential by use and input position	Consumer, slot, invocation	Component input map	Same signed differential for the declared branch or global cut	Evaluated separately

S4.1 Bilingual terminology contract

Chinese term	English term	Definition boundary
结构合法切面	structurally valid cut	Every structural input-to-target use-path crosses exactly once; independent of current values
点态微分闭合	pointwise differential closure	Sum of use-specific Jacobian/cotangent terms at a fixed input
分支微分分组	branch differential partition	Reconstructs only an attention or MLP branch contribution
全局切面	global cut	Includes bypass and reconstructs same-target Total Endpoint
路径图	path map	Local differential term for a specified tensor use, not a unique causal path
局部微分绝对质量比	local-differential absolute-mass ratio	Routing/identity scale; not a causal contribution fraction
区域诱导的指定-head activation 替换	region-induced replacement of designated head activations	Two-stage task: obtain a head activation from a mixed input, then insert it into a clean forward
候选集合遗憾率	candidate-set regret	Relative to the evaluated candidate-set envelope, not a globally optimal policy

S5. Corrected absolute random AUC and oracle

The current random baseline takes absolute effects before integration within each repeat, then averages AUCs. Corrected results, repeat distributions and MC stability are in supplementary experiments S9. The old analysis is retained

as a historical reconstruction of its original estimand.

S6. Detailed attention/residual evidence (historical stage identifiers retained)

Historical M4: attention routing/value and residual mechanisms

M4 covered a shared 400-image cohort, both models, all 12 blocks, and every head for the frozen ground-truth logit. The result contains 86,400 image-block-head attention records, 19,200 image-block-residual records, and 691,200 direction-head JVP records. Per-head Q+K+V reconstructed the native attention Endpoint, routing+value reconstructed the same object, and identity/carry plus branch reconstructed the full logit Endpoint at both residual additions.

The independent CPU audit recomputed 230,400 E3 summary fields. Maximum absolute Q/K/V and residual closure errors were 1.574e-05 and 2.480e-05. The maximum relative error between the explicit attention differential and native JVP was 2.502e-05. All 19,200 identity/carry omission controls failed closure as intended, and 384 attention-detach block checks passed for the first 16 frozen evaluation orders.

Attention routing and value

Model	Block	Routing share	Frozen-attention Endpoint error	Frozen-attention JVP error	Mean-query saturation	Routing-value path cancellation
vit_b16	0	0.3092	0.4407	0.4508	0.4656	0.1555
vit_b16	5	0.5239	0.7792	0.8547	0.3183	0.2342
vit_b16	11	0.5154	0.8022	0.9516	0.3470	0.3069
deit_s16	0	0.2041	0.2774	0.3099	0.3825	0.1377
deit_s16	5	0.4852	0.7375	0.8548	0.3191	0.2635
deit_s16	11	0.4808	0.8488	0.9636	0.3329	0.3692

Frozen attention retains only A dV. Endpoint and JVP errors therefore quantify omitted routing differentials along the input-scaling direction and eight frozen Rademacher directions, respectively. The tests below aggregate heads and blocks within each image before image-level inference; Holm families were frozen separately by model.

Model	Domain	Type	Frozen test	Image-level median	Pos/neg/tie	Holm p	Decision
vit_b16	attention	attention-head	late_minus_early_routing_share	0.0523	360/40/0	1.715e-65	SUPPORTED_DIFFERENCE
vit_b16	attention	attention-head	late_minus_early_frozen_attention_endpoint_error	0.1441	383/17/0	5.545e-91	SUPPORTED_DIFFERENCE
vit_b16	attention	attention-head	saturation_routing_relation	-0.2937	5/395/0	2.611e-109	SUPPORTED_DIFFERENCE
vit_b16	attention	attention-head	saturation_error_relation	-0.2238	12/388/0	7.108e-98	SUPPORTED_DIFFERENCE
vit_b16	residual	attention	late_minus_early_identity_share	-0.1242	38/362/0	4.068e-67	SUPPORTED_DIFFERENCE
vit_b16	residual	attention	late_minus_early_path_cancellation	-0.0105	173/227/0	7.970e-03	SUPPORTED_DIFFERENCE
vit_b16	residual	mlp	late_minus_early_identity_share	0.1798	396/4/0	1.644e-111	SUPPORTED_DIFFERENCE
vit_b16	residual	mlp	late_minus_early_path_cancellation	0.0688	304/96/0	2.840e-26	SUPPORTED_DIFFERENCE
deit_s16	attention	attention-head	late_minus_early_routing_share	0.1355	399/1/0	9.317e-118	SUPPORTED_DIFFERENCE
deit_s16	attention	attention-head	late_minus_early_frozen_attention_endpoint_error	0.2896	400/0/0	3.098e-120	SUPPORTED_DIFFERENCE
deit_s16	attention	attention-head	saturation_routing_relation	-0.3143	15/379/6	5.158e-92	SUPPORTED_DIFFERENCE
deit_s16	attention	attention-head	saturation_error_relation	-0.2286	52/338/10	1.794e-52	SUPPORTED_DIFFERENCE
deit_s16	residual	attention	late_minus_early_identity_share	-0.0449	48/352/0	6.053e-58	SUPPORTED_DIFFERENCE
deit_s16	residual	attention	late_minus_early_path_cancellation	-0.0074	171/229/0	4.314e-03	SUPPORTED_DIFFERENCE
deit_s16	residual	mlp	late_minus_early_identity_share	0.1780	397/3/0	1.652e-113	SUPPORTED_DIFFERENCE
deit_s16	residual	mlp	late_minus_early_path_cancellation	0.0588	103/297/0	6.661e-23	SUPPORTED_DIFFERENCE

Saturation relations use Spearman correlations across heads within each image and layer before aggregation, separating head-level association from common depth trends. The table reports their observed directions without assuming that sharper softmax rows must yield larger routing omission.

Residual identity and branch

Model	Residual	Block	Identity share	Path cancellation	Identity-branch cosine	Output-identity patch TV
vit_b16	attention	0	0.8847	0.1465	-0.3315	0.0150
vit_b16	attention	5	0.6100	0.2381	0.0606	0.1089
vit_b16	attention	11	0.4442	0.3578	-0.0506	0.1803
vit_b16	mlp	0	0.2225	0.1654	0.1849	0.1398
vit_b16	mlp	5	0.5648	0.2450	0.1321	0.1287
vit_b16	mlp	11	0.6418	0.6333	-0.9071	0.1023
deit_s16	attention	0	0.8729	0.1820	-0.4181	0.0284
deit_s16	attention	5	0.6775	0.1983	0.1826	0.0798
deit_s16	attention	11	0.6559	0.2538	0.0757	0.1047
deit_s16	mlp	0	0.3078	0.2831	-0.0186	0.1798
deit_s16	mlp	5	0.6257	0.1886	0.3342	0.0923
deit_s16	mlp	11	0.6079	0.5878	-0.7599	0.1276

Identity share partitions absolute Endpoint mass between the skip and branch paths. Path cancellation compares their signed sum with the sum of their absolute masses. Output-identity patch TV measures the spatial change from the identity-only distribution to the residual output. For ViT-B/16, the minimum/maximum spatial-TV blocks were 0/11 for attention residuals and 8/10 for MLP residuals; the DeiT-S pairs were 1/4 and 9/0. These locations are descriptive medians over the frozen cohort.

Model	Phase	Attention−MLP identity share	Attention−MLP path cancellation	Attention−MLP spatial TV
vit_b16	early	0.3382	0.0004	-0.0902
vit_b16	middle	0.0327	0.0165	-0.0050
vit_b16	late	0.0376	-0.0664	-0.0173
deit_s16	early	0.2697	-0.0564	-0.0796
deit_s16	middle	0.0280	0.0009	-0.0026
deit_s16	late	0.0512	-0.0041	-0.0047

Model	Residual	Branch share–path cancellation Spearman across 12 blocks
vit_b16	attention	0.5140
vit_b16	mlp	-0.2657
deit_s16	attention	0.3182
deit_s16	mlp	0.4650

The first table directly compares attention and MLP residuals. The second asks whether path cancellation changes with branch mass. Both are frozen A1 descriptive analyses outside the original M4 Holm families.

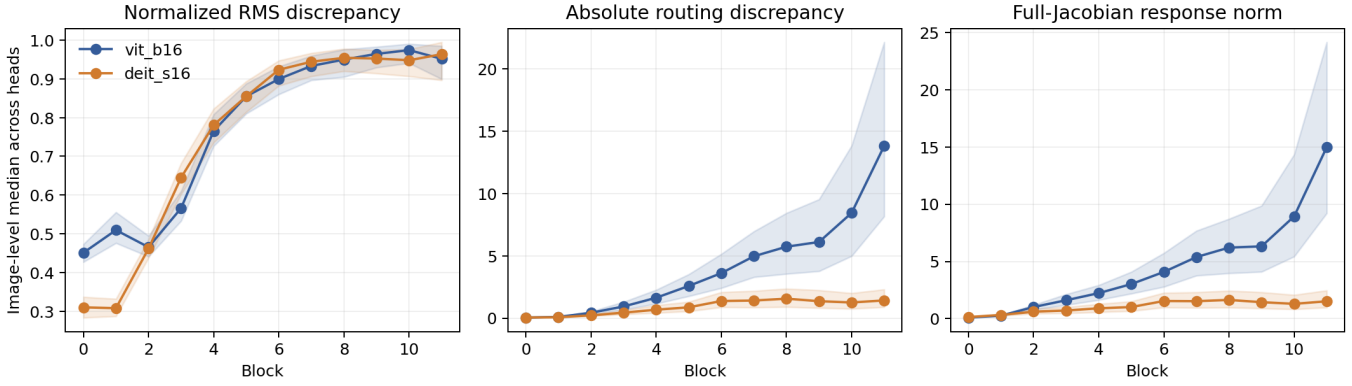
Signed, absolute, and cancellation views

Every path retains signed sum, absolute mass, positive/negative mass, and within-map cancellation. M4 keeps within-map cancellation separate from routing/value or identity/branch path cancellation. Zero denominators, zero norms, and constant-head correlations propagate as NA without epsilon substitution.

Model	Path	Block	Signed magnitude	Absolute mass	Map cancellation
vit_b16	routing	0	0.0362	2.6596	0.9867
vit_b16	routing	5	0.0600	33.7856	0.9982
vit_b16	routing	11	0.0608	40.7842	0.9982
vit_b16	value	0	0.0306	7.6041	0.9952
vit_b16	value	5	0.0380	28.0522	0.9986
vit_b16	value	11	0.0654	39.0241	0.9981
vit_b16	attention_identity	0	1.3315	495.2271	0.9972
vit_b16	attention_identity	5	0.9816	387.1208	0.9977
vit_b16	attention_identity	11	0.8778	313.3689	0.9974
vit_b16	attention_branch	0	0.3930	63.8344	0.9934
vit_b16	attention_branch	5	0.4140	250.3045	0.9983
vit_b16	attention_branch	11	0.6133	398.0202	0.9986
vit_b16	mlp_identity	0	0.6860	130.4998	0.9945
vit_b16	mlp_identity	5	1.0976	334.0699	0.9971
vit_b16	mlp_identity	11	1.8214	827.8407	0.9977
vit_b16	mlp_branch	0	0.6309	438.0538	0.9986
vit_b16	mlp_branch	5	0.4504	271.9139	0.9983
vit_b16	mlp_branch	11	0.7961	460.2701	0.9981
deit_s16	routing	0	0.0756	4.5294	0.9829
deit_s16	routing	5	0.0506	38.3865	0.9986
deit_s16	routing	11	0.0504	35.9257	0.9986
deit_s16	value	0	0.0818	18.4261	0.9950
deit_s16	value	5	0.0776	40.1955	0.9979
deit_s16	value	11	0.0586	38.8767	0.9985
deit_s16	attention_identity	0	1.6888	459.3077	0.9960
deit_s16	attention_identity	5	1.1087	370.3635	0.9969
deit_s16	attention_identity	11	1.1539	372.6598	0.9966
deit_s16	attention_branch	0	0.4543	63.9204	0.9922
deit_s16	attention_branch	5	0.4112	169.7186	0.9974
deit_s16	attention_branch	11	0.3639	185.3893	0.9980
deit_s16	mlp_identity	0	0.8741	170.6143	0.9946
deit_s16	mlp_identity	5	1.1847	317.7498	0.9962
deit_s16	mlp_identity	11	1.7620	614.5721	0.9973
deit_s16	mlp_branch	0	0.7619	412.5634	0.9981
deit_s16	mlp_branch	5	0.3466	197.6662	0.9982
deit_s16	mlp_branch	11	0.4532	354.6503	0.9988

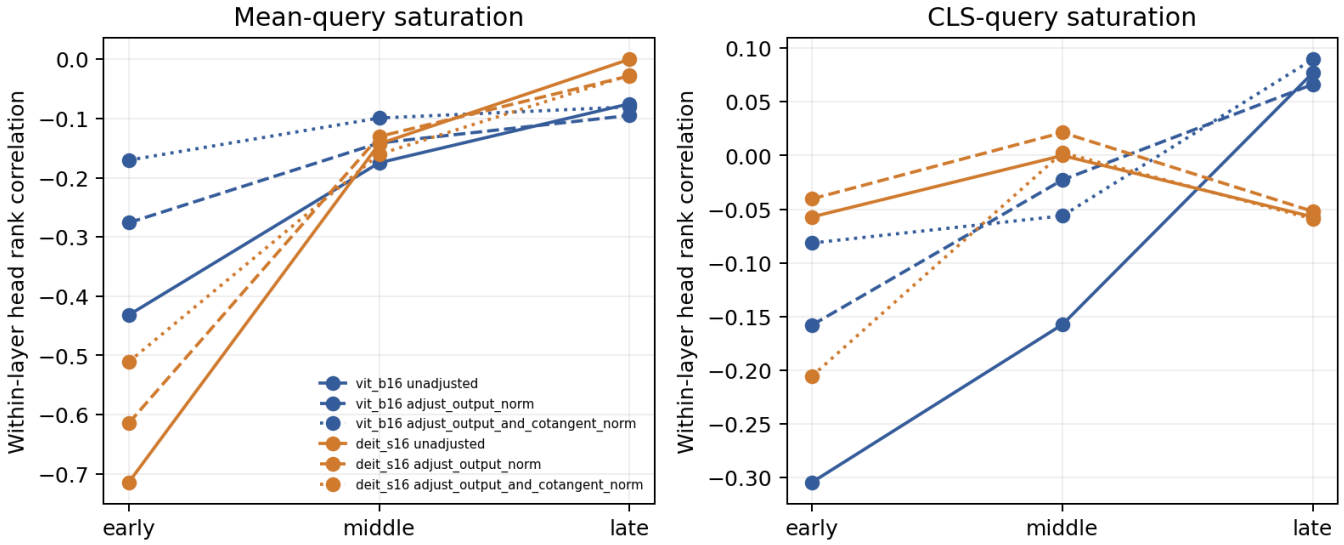
Review-R2 sensitivity and estimand refinement

Review-R2 tested whether the large late-block Value-only JVP discrepancies were driven by small denominators. At block 11, the median relative discrepancies were 0.9516 for ViT-B/16 and 0.9636 for DeiT-S/16; the corresponding full-response denominators were 14.9711 and 1.5155, while absolute discrepancies were 13.8064 and 1.4228. The discrepancies therefore occur at the same norm scale as the full response rather than arising generally from near-zero denominators.



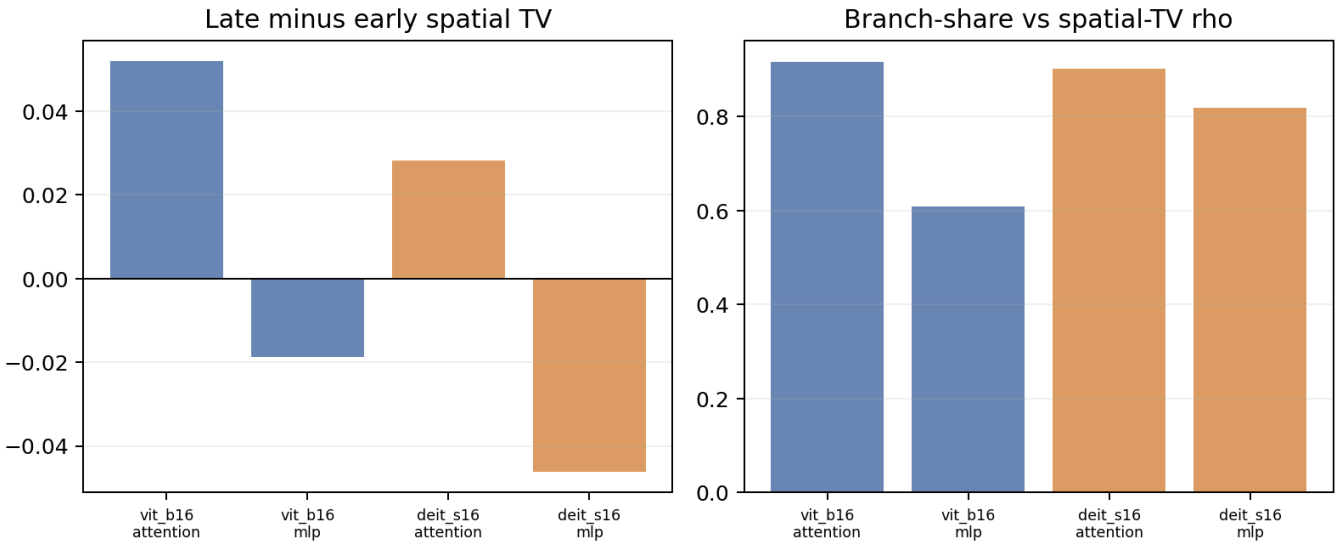
M4 Review Archived R2 Figure 1. Block-11 distributions of normalized JVP discrepancy, absolute discrepancy, and full-response denominator for both models.

After adjustment for head projection-input activation norm and downstream-cotangent norm, mean-query saturation remained negatively associated with routing share, Endpoint error, and JVP error in both models. The fully adjusted correlations were −0.1442/−0.3257, −0.1129/−0.2185, and −0.1453/−0.2786 for ViT-B/16 and DeiT-S/16, respectively. CLS-query relations were weaker: the adjusted ViT-B routing-share and Endpoint-error tests had Holm $p = 0.0629$. The supported summary therefore concerns all-query mean saturation, with the association concentrated in early and middle blocks.



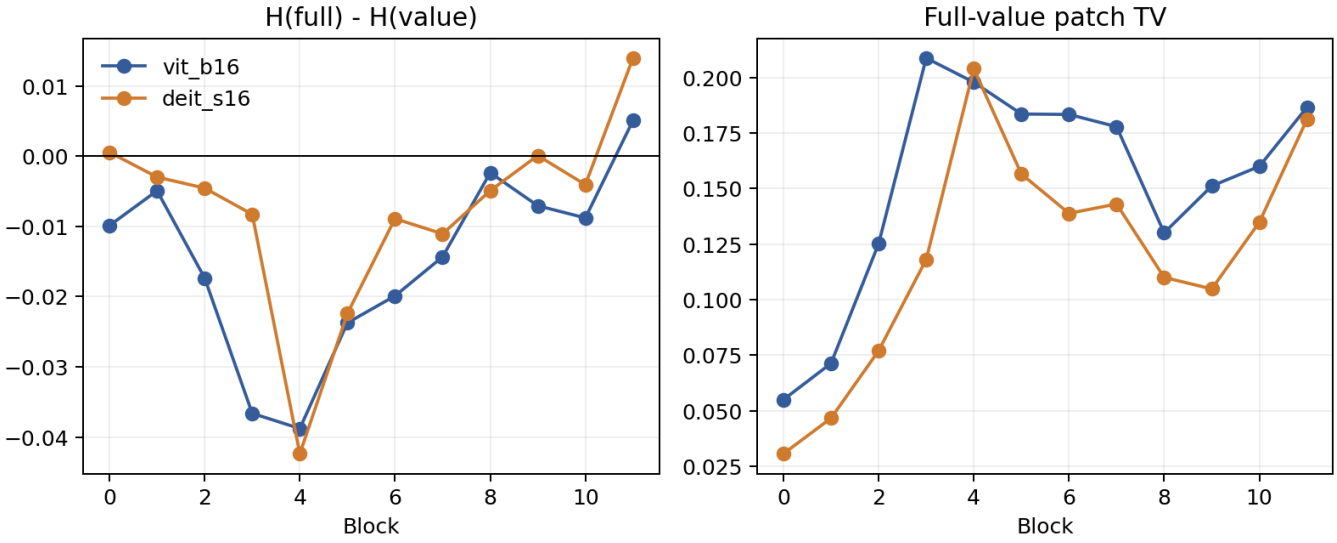
M4 Review Archived R2 Figure 2. Unadjusted and partial-rank saturation relations, including phase-level sensitivity and the weaker CLS-query result.

The full-cohort spatial analysis measured total variation between output and identity 14×14 patch absolute distributions. Attention residual late-minus-early differences were +0.0520 ([0.0488, 0.0547]) for ViT-B/16 and +0.0282 ([0.0257, 0.0300]) for DeiT-S/16; MLP residual differences were −0.0187 ([−0.0220, −0.0150]) and −0.0462 ([−0.0516, −0.0412]). All four post-hoc sensitivity tests survived Holm correction. Branch-relative path scale and spatial TV were positively associated within images across blocks.



M4 Review Archived R2 Figure 3. Full-cohort residual spatial-change sensitivity for attention and MLP sublayers.

On 16 pre-retained class-logit maps per model, Full–Value patch TV rose from 0.0550/0.0307 at block 0 to 0.1865/0.1811 at block 11. Its 12-block trajectory was directionally compatible with M3 entropy, locality, and distance, but M3 used 100-image token-channel targets with only one overlapping image per model. This is a model-level trajectory bridge, not a unit-level or causal link between routing and spatial broadening.



M4 Review Archived R2 Figure 4. Descriptive bridge between M3 spatial trajectories and full-versus-Value class-logit map differences.

Figures

- Image #1 (m4_attention_depth): routing share, frozen-attention Endpoint error, and eight-direction JVP error across blocks; bands are 10,000-replicate image-bootstrap 95% intervals.
- Image #2 (m4_attention_saturation): descriptive block-level locations of saturation versus routing/error; formal association tests use within-image, within-layer head Spearman correlations.

- Image #3 (m4_residual_depth): identity share, path cancellation, and spatial TV for attention and MLP residuals.

Interpretation boundary

M4 quantifies local Jacobian path accounting at the observed input for the frozen ground-truth logit. A larger routing share or branch mass indicates more local input dependence along that path; causal importance requires separate finite-intervention evidence.

S7. Complete supplementary image index

- S01 — S01_figure_01_endpoint_scanner_method_overview
- S02 — S02_figure_02_endpoint_cnn_and_path_method_identity
- S03 — S03_figure_03_same_target_endpoint_versus_path
- S04 — S04_figure_04_original_cnn_endpoint_gates
- S05 — S05_figure_05_attention_q_k_v_differential_paths
- S06 — S06_figure_06_residual_edge_accounting
- S07 — S07_figure_07_unit_sum_and_energy_targets
- S08 — S08_figure_08_state_seed_and_input_coordinates
- S09 — S09_figure_09_layernorm_and_gelu_jacobians
- S10 — S10_figure_10_structured_edge_cuts
- S11 — S11_figure_11_registered_target_atlas_design
- S12 — S12_figure_12_sloth_bear_class_logit_maps_across_models
- S13 — S13_figure_13_sloth_bear_multi_target_atlas
- S14 — S14_figure_14_scan_scale_latency_and_resources
- S15 — S15_figure_15_structural_closure_contracts
- S16 — S16_figure_16_formal_closure_and_negative_controls
- S17 — S17_figure_17_sloth_bear_structural_component_maps
- S18 — S18_figure_18_condition_scaled_analytic_jvp_checks
- S19 — S19_figure_19_matched_finite_activation_replacement
- S20 — S20_figure_20_c4a_family_estimates_and_intervals
- S21 — S21_figure_21_rq3_threshold_zoom_and_boundary_exception
- S22 — S22_figure_22_endpoint_and_registered_secondary_predictors
- S23 — S23_figure_23_spatial_intervention_protocol
- S24 — S24_figure_24_sloth_bear_block_5_maps_for_spatial_ranking
- S25 — S25_figure_25_one_image_spatial_effect_curves
- S26 — S26_figure_26_c4b_mandatory_baseline_contrasts
- S27 — S27_figure_27_rq4_family_summary
- S28 — S28_figure_28_identity_correctness_utility_conclusion
- S29 — S29_figure_29_m3_block_trajectories
- S30 — S30_figure_30_m3_cls_minus_patch_entropy_by_block
- S31 — S31_figure_31_m3_cancellation_scales
- S32 — S32_figure_32_original_nns_versus_endpoint_numerical_equivalence

- S33 — S33_figure_33_m3_finite_difference_diagnostics
- S34 — S34_figure_34_registered_target_learning_images
- S35 — S35_figure_35_block5_structural_learning_images
- S36 — S36_figure_36_spatial_ranking_learning_images
- S37 — S37_figure_37_channel0_depth_progression
- S38 — S38_figure_38_attention_head_depth_atlas
- S39 — S39_figure_39_vit_b16_attention_head_depth_atlas
- S40 — S40_figure_40_vit_b16_module_chain
- S41 — S41_figure_12_sloth_bear_class_logit_maps_across_models_signed
- S42 — S42_figure_13_sloth_bear_multi_target_atlas_signed
- S43 — S43_figure_17_sloth_bear_structural_component_maps_signed
- S44 — S44_figure_24_sloth_bear_block_5_maps_for_spatial_ranking_signed
- S45 — S45_block_00
- S46 — S46_block_01
- S47 — S47_block_02
- S48 — S48_block_03
- S49 — S49_block_04
- S50 — S50_block_05
- S51 — S51_block_06
- S52 — S52_block_07
- S53 — S53_block_08
- S54 — S54_block_09
- S55 — S55_block_10
- S56 — S56_block_11
- S57 — S57_block_00
- S58 — S58_block_01
- S59 — S59_block_02
- S60 — S60_block_03
- S61 — S61_block_04
- S62 — S62_block_05
- S63 — S63_block_06
- S64 — S64_block_07
- S65 — S65_block_08
- S66 — S66_block_09
- S67 — S67_block_10
- S68 — S68_block_11
- S69 — S69_supp_m2_01_M2_F1
- S70 — S70_supp_m2_02_M2_F2
- S71 — S71_supp_m2_03_M2_F3
- S72 — S72_supp_m3_01_m3_atlas_depth
- S73 — S73_supp_m3_02_m3_map_similarity
- S74 — S74_supp_m4_01_m4_attention_depth
- S75 — S75_supp_m4_02_m4_attention_saturation
- S76 — S76_supp_m4_03_m4_residual_depth

- S77 — S77_supp_m4_review_01_m4_review_jvp_diagnostics
- S78 — S78_supp_m4_review_02_m4_review_saturation_robustness
- S79 — S79_supp_m4_review_03_m4_review_residual_spatial
- S80 — S80_supp_m4_review_04_m4_review_m3_m4_bridge
- S81 — S81_supp_m6_01_m6_neuron_atlas
- S82 — S82_supp_m6_02_m6_model_layer_comparison
- S83 — S83_supp_m6_03_m6_attention_cases
- S84 — S84_supp_m6_04_m6_residual_cases
- S85 — S85_supp_m6_05_m6_m2_semantic_boundary
- S86 — S86_figure_34_registered_target_learning_images_signed
- S87 — S87_figure_35_block5_structural_learning_images_signed
- S88 — S88_figure_36_spatial_ranking_learning_images_signed
- S89 — S89_figure_37_channel0_depth_progression_signed
- S90 — S90_figure_38_attention_head_depth_atlas_signed
- S91 — S91_figure_39_vit_b16_attention_head_depth_atlas_signed
- S92 — S92_figure_40_vit_b16_module_chain_signed

S8. Current qualitative comparisons

The six CPU image-model computations use the same source images, checkpoints and preprocessing. Grad-CAM uses the last norm1 patch activations; attention rollout and raw attention are class-agnostic. These presentation-selected examples add no population tests. See display protocol and compute receipt.

Revision v2 moves the following presentation-oriented or repetitive figures out of the main text. File names retain their original publication-R2 numbers to preserve hashes and provenance bindings:

- Coordinate initialization
- Local-derivative propagation
- Patch embedding
- DeiT-S/16 across twelve layers
- ViT-B/16 across twelve layers
- Module-level learning images and forward activations
- All attention heads
- Cross-model display examples
- DeiT-S/16 method display
- ViT-B/16 method display
- Positive, signed, and absolute reading views

Table S6. Cross-reference rank concordance.

Model	Quantity	Spearman	Kendall tau-b	Top-1 agreement	Top-3 overlap	Mean abs rank displacement
DeiT-S/16	finite_effect	0.3071	0.2444	0.4167	0.6111	1.5278
DeiT-S/16	taylor	0.3000	0.2389	0.3333	0.6111	1.5278
ViT-B/16	finite_effect	0.3759	0.2803	0.3333	0.5000	2.9444
ViT-B/16	taylor	0.3782	0.2828	0.3333	0.5000	2.9444

New supplementary experiments S9–S14

This supplement executes the minimum package in Section 11 of the experiment request and both P0 corrections. New estimates are supplementary descriptive analyses; original confirmatory decisions and sealed evidence are retained. Cohorts, channels, initialization seeds and estimators were fixed before model execution.

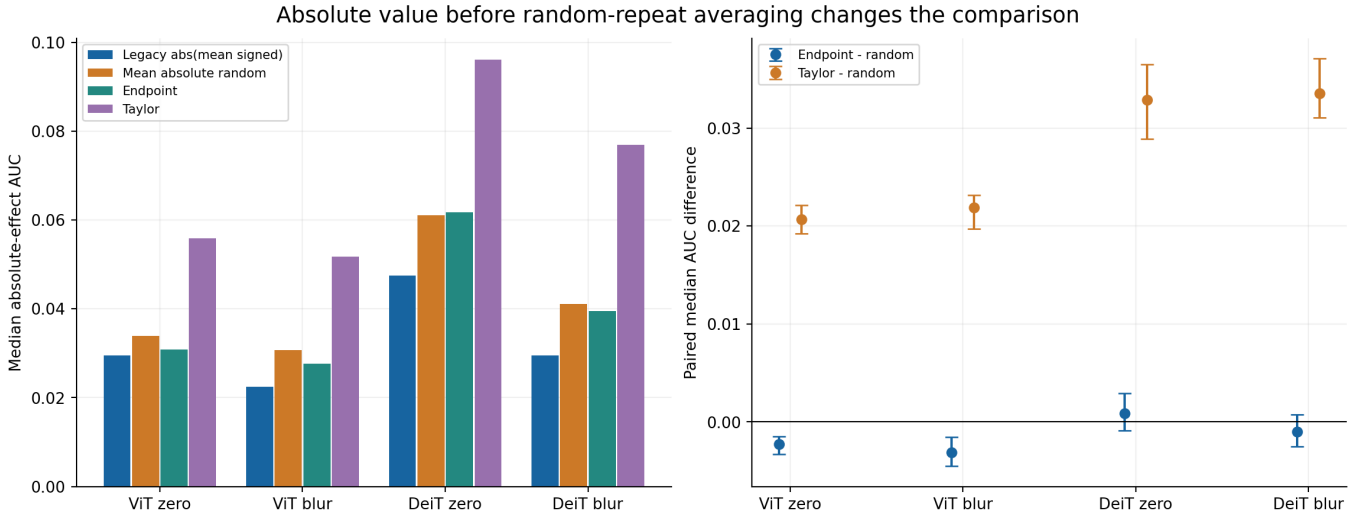
S9. Correction of the random-region baseline

Each random repetition takes the absolute effect before AUC integration; its AUC is then averaged over 20 repetitions and heads within an image. The legacy estimator averaged signed curves first and underestimated absolute-effect random performance through cancellation. The table reports medians and paired median differences for 400-image families, with 2,000 image-bootstrap intervals.

$$\widehat{A}_{\text{random}} = \frac{1}{R}\sum_{r=1}^{R}\frac{1}{0.25}\int_{0}^{0.25}|e_r(a)|\,da. \tag{1}$$

Model	Reference	Legacy random median	Corrected random mean	Corrected random median	Endpoint - random median [95% CI]	Taylor - random median [95% CI]
ViT-B/16	normalized-zero	0.0294	0.0366	0.0339	-0.0023 [-0.0033, -0.0015]	0.0207 [0.0192, 0.0221]
ViT-B/16	blur	0.0225	0.0355	0.0307	-0.0031 [-0.0045, -0.0016]	0.0219 [0.0197, 0.0232]
DeiT-S/16	normalized-zero	0.0475	0.0728	0.0611	0.0009 [-0.0009, 0.0029]	0.0329 [0.0289, 0.0365]
DeiT-S/16	blur	0.0296	0.0505	0.0411	-0.0010 [-0.0025, 0.0007]	0.0336 [0.0311, 0.0371]

After correction, both ViT-B Endpoint-minus-random median differences are negative; both DeiT-S intervals include zero. All four Taylor-minus-random intervals remain positive. The Endpoint–Taylor comparison is unchanged.



The median per-image relative Monte Carlo SE is 5.1%–6.9%, with q90 of 7.6%–8.9%. Resampling 5/10/20 repetitions from the observed pool shows increasing error at smaller repetition counts. Conditional on the fixed images, cohort-mean Monte Carlo variation is smaller than image-sampling uncertainty; small single-image advantages remain poorly

resolved. This diagnostic covers the observed repetition pool and does not establish that 20 repetitions meet every precision requirement.

Model	Reference	Median relative MC SE	q90 relative MC SE	MC image error R=5	R=10	R=20
ViT-B/16	normalized-zero	0.0506	0.0759	0.0020	0.0014	0.0010
ViT-B/16	blur	0.0665	0.0883	0.0026	0.0018	0.0013
DeiT-S/16	normalized-zero	0.0622	0.0851	0.0046	0.0032	0.0023
DeiT-S/16	blur	0.0687	0.0895	0.0035	0.0024	0.0017

20-repeat cohort distribution · Per-image repeats · Paired image values, MC SE, split-half and leave-one-repeat-out

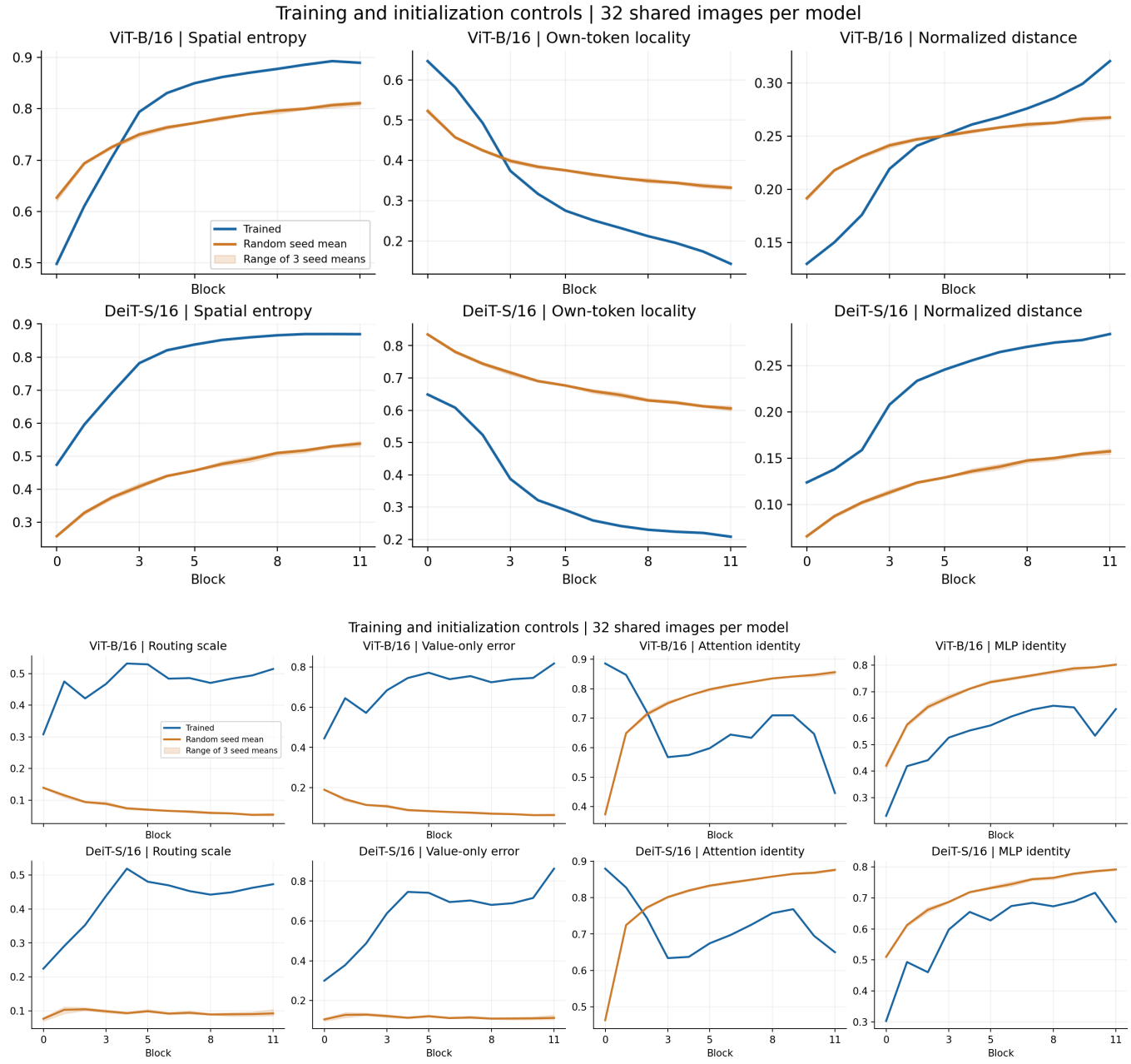
S10. Trained and randomly initialized models

Each ViT-B/16 and DeiT-S/16 architecture has three random states (202609201, 202609202, 202609203), sharing the first 32 frozen M3 images, all 12 blocks, 16 tokens and 16 frozen channels per block with the trained model; M4 includes all heads. Random states use the same timm 0.9.16 architecture and native initialization, and each state is fixed across images. Trained statistics are selected from existing M3/M4 raw arrays for the identical images and targets. Units are aggregated within images; early=0–3 and late=8–11. Trained-minus-random intervals independently resample shared images and initialization seeds in 2,000 crossed-bootstrap draws.

Random models already show increasing entropy and distance and decreasing locality, so spatial broadening is not unique to the trained parameter state. Trained entropy late-minus-early is 0.2288/0.2256, compared with random-state means of 0.0940/0.1718; the two parameter states differ in broadening magnitude. Routing scale increases in the trained parameter states while its random-state mean trend is negative, and attention identity changes in opposite directions between the two states. MLP identity increases in both, with the ViT-B trained-state-minus-random-state interval including zero. Three random states limit inference over the initialization population; this comparison does not identify a causal effect of training or separately identify architecture and initialization effects.

Model	Study	Metric	Trained late - early	Random late - early	Trained - random [95% CI]	Random seed 1 / 2 / 3
DeiT-S/16	m3	cancellation	0.0020	0.0036	-0.0016 [-0.0029, -0.0005]	0.0032 / 0.0028 / 0.0050
DeiT-S/16	m3	cls__minus__patch__energy	-0.2685	-0.1717	-0.0968 [-0.1051, -0.0889]	-0.1697 / -0.1740 / -0.1714
DeiT-S/16	m3	distance	0.1276	0.0576	0.0700 [0.0624, 0.0781]	0.0574 / 0.0584 / 0.0569
DeiT-S/16	m3	entropy	0.2256	0.1718	0.0539 [0.0473, 0.0602]	0.1699 / 0.1740 / 0.1714
DeiT-S/16	m3	locality	-0.3441	-0.1445	-0.1997 [-0.2115, -0.1882]	-0.1433 / -0.1466 / -0.1436
DeiT-S/16	m4	attention__identity__scale	0.0579	0.1188	-0.1767 [-0.1916, -0.1618]	0.1232 / 0.1142 / 0.1190
DeiT-S/16	m4	attention__output__identity_tv	0.0306	-0.0026	0.0331 [0.0274, 0.0388]	-0.0029 / -0.0025 / -0.0023
DeiT-S/16	m4	attention__path__cancellation	0.0958	-0.0042	0.1001 [0.0890, 0.1115]	0.0002 / -0.0036 / -0.0093
DeiT-S/16	m4	attention__residual__path__cancellation	0.0188	-0.0848	0.1036 [0.0810, 0.1271]	-0.0889 / -0.0770 / -0.0884
DeiT-S/16	m4	jvp__discrepancy	0.5465	-0.0944	0.6409 [0.6150, 0.6673]	-0.0871 / -0.0859 / -0.1101
DeiT-S/16	m4	mlp__identity__scale	0.2063	0.1461	0.0602 [0.0308, 0.0907]	0.1423 / 0.1431 / 0.1528

Model	Study	Metric	Trained late - early	Random late - early	Trained - random [95% CI]	Random seed 1 / 2 / 3
DeiT-S/16	m4	mlp__output__identity_tv	-0.0610	-0.0041	-0.0569 [-0.0722, -0.0413]	-0.0047 / -0.0042 / -0.0034
DeiT-S/16	m4	mlp__residual__path__cancellation	-0.0381	-0.0756	0.0375 [0.0101, 0.0675]	-0.0731 / -0.0761 / -0.0775
DeiT-S/16	m4	routing__scale	0.1314	-0.0043	0.1357 [0.1151, 0.1564]	0.0023 / -0.0032 / -0.0120
DeiT-S/16	m4	saturation__error	0.7536	-0.0348	0.7884 [0.6387, 0.9354]	-0.0348 / -0.0348 / -0.0348
DeiT-S/16	m4	saturation__routing	0.7598	-0.0348	0.7946 [0.6417, 0.9491]	-0.0357 / -0.0339 / -0.0348
DeiT-S/16	m4	value__error	0.2676	-0.0088	0.2764 [0.2406, 0.3122]	-0.0008 / -0.0071 / -0.0184
ViT-B/16	m3	cancellation	0.0024	0.0008	0.0016 [0.0008, 0.0023]	0.0014 / 0.0010 / 0.0000
ViT-B/16	m3	cls__minus__patch__entropy	-0.2541	-0.0940	-0.1802 [-0.1893, -0.1709]	-0.0933 / -0.0970 / -0.0917
ViT-B/16	m3	distance	0.1292	0.0399	0.0893 [0.0833, 0.0953]	0.0399 / 0.0410 / 0.0387
ViT-B/16	m3	entropy	0.2288	0.0940	0.1348 [0.1232, 0.1466]	0.0934 / 0.0968 / 0.0918
ViT-B/16	m3	locality	-0.3522	-0.1005	-0.2517 [-0.2598, -0.2431]	-0.1001 / -0.1036 / -0.0979
ViT-B/16	m4	attention__identity__scale	-0.1144	0.1631	-0.2775 [-0.3004, -0.2520]	0.1629 / 0.1662 / 0.1603
ViT-B/16	m4	attention__output__identity_tv	0.0507	-0.0046	0.0553 [0.0454, 0.0644]	-0.0042 / -0.0040 / -0.0058
ViT-B/16	m4	attention__path__cancellation	0.0797	-0.0421	0.1218 [0.1136, 0.1303]	-0.0428 / -0.0413 / -0.0423
ViT-B/16	m4	attention__residual__path__cancellation	0.0138	-0.1050	0.1188 [0.0932, 0.1447]	-0.1028 / -0.1084 / -0.1038
ViT-B/16	m4	jvp__discrepancy	0.4501	-0.2600	0.7101 [0.6739, 0.7465]	-0.2621 / -0.2548 / -0.2631
ViT-B/16	m4	mlp__identity__scale	0.1951	0.1819	0.0132 [-0.0093, 0.0361]	0.1829 / 0.1851 / 0.1775
ViT-B/16	m4	mlp__output__identity_tv	-0.0205	-0.0042	-0.0163 [-0.0295, -0.0047]	-0.0041 / -0.0045 / -0.0041
ViT-B/16	m4	mlp__residual__path__cancellation	0.0887	-0.0866	0.1753 [0.1350, 0.2174]	-0.0845 / -0.0907 / -0.0846
ViT-B/16	m4	routing__scale	0.0512	-0.0479	0.0991 [0.0810, 0.1160]	-0.0485 / -0.0470 / -0.0481
ViT-B/16	m4	saturation__error	0.3568	0.0232	0.3336 [0.2370, 0.4496]	0.0827 / -0.0498 / 0.0366
ViT-B/16	m4	saturation__routing	0.4379	0.0203	0.4177 [0.3136, 0.5211]	0.0636 / -0.0431 / 0.0402
ViT-B/16	m4	value__error	0.1437	-0.0623	0.2061 [0.1689, 0.2420]	-0.0621 / -0.0615 / -0.0633



Early/late and difference estimates · Image values

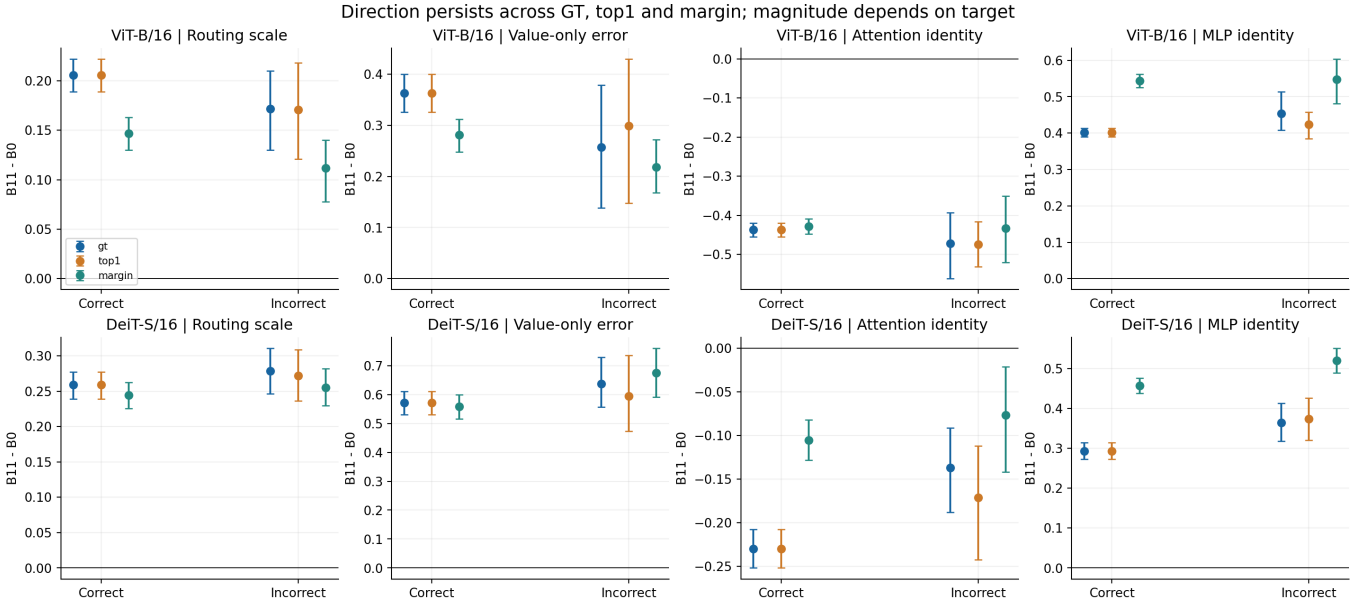
S11. Target robustness

The frozen M3 cohort supplies 100 images per model, with GT, clean-top1 and top1-minus-runnerup targets at blocks 0/5/11 and all heads. Class indices are fixed on the clean input. Correct/incorrect counts are 92/8 for ViT-B and 86/14 for DeiT-S; GT and top1 are identical on correct images. The table reports image-level B11-minus-B0 after head aggregation for all/correct/incorrect groups. Targets within a group share bootstrap image indices.

Routing scale and Value-only error increase with depth under all three targets; attention identity decreases and MLP identity increases, with consistent directions in correct and incorrect subsets. Margin targets change magnitudes: ViT-B routing increases by 0.1439 rather than the GT value of 0.2031, and DeiT-S attention identity changes by

−0.1014 rather than −0.2168. The incorrect subsets contain only 8/14 images, limiting their precision.

Native attention-output JVPs depend on the model and input, independently of the explained class, so their eight-direction discrepancy is inherited from the identical original M4 images. The additional target_contracted_jvp contracts directional derivatives with each target cotangent at the head output and measures target-dependent Value-only discrepancy. The two quantities are reported separately. Saturation relations use within-block Spearman across heads; their depth changes are reported alongside complete block values, target contrasts and NA counts in the data files.



Model	Subset	Target	Metric	n	B11 - B0 mean	95% CI
ViT-B/16	all	gt	routing_scale	100	0.2031	[0.1868, 0.2189]
ViT-B/16	all	top1	routing_scale	100	0.2030	[0.1862, 0.2184]
ViT-B/16	all	margin	routing_scale	100	0.1439	[0.1283, 0.1595]
ViT-B/16	all	gt	value_error	100	0.3539	[0.3170, 0.3877]
ViT-B/16	all	top1	value_error	100	0.3572	[0.3203, 0.3917]
ViT-B/16	all	margin	value_error	100	0.2762	[0.2448, 0.3059]
ViT-B/16	all	gt	jvp_discrepancy	100	0.4757	[0.4569, 0.4942]
ViT-B/16	all	top1	jvp_discrepancy	100	0.4757	[0.4569, 0.4942]
ViT-B/16	all	margin	jvp_discrepancy	100	0.4757	[0.4569, 0.4942]
ViT-B/16	all	gt	attention_path_cancellation	100	0.1587	[0.1469, 0.1699]
ViT-B/16	all	top1	attention_path_cancellation	100	0.1601	[0.1488, 0.1713]
ViT-B/16	all	margin	attention_path_cancellation	100	0.1427	[0.1326, 0.1529]
ViT-B/16	all	gt	saturation_routing	100	-0.1076	[-0.1852, -0.0322]
ViT-B/16	all	top1	saturation_routing	100	-0.1087	[-0.1849, -0.0364]
ViT-B/16	all	margin	saturation_routing	100	0.0102	[-0.0637, 0.0853]
ViT-B/16	all	gt	saturation_error	100	-0.0495	[-0.1332, 0.0347]
ViT-B/16	all	top1	saturation_error	100	-0.0517	[-0.1335, 0.0312]
ViT-B/16	all	margin	saturation_error	100	0.0421	[-0.0353, 0.1223]
ViT-B/16	all	gt	target_contracted_jvp	100	0.3382	[0.3029, 0.3751]
ViT-B/16	all	top1	target_contracted_jvp	100	0.3414	[0.3046, 0.3786]
ViT-B/16	all	margin	target_contracted_jvp	100	0.2887	[0.2568, 0.3201]
ViT-B/16	all	gt	attention_identity_scale	100	-0.4390	[-0.4577, -0.4213]
ViT-B/16	all	top1	attention_identity_scale	100	-0.4392	[-0.4568, -0.4219]
ViT-B/16	all	margin	attention_identity_scale	100	-0.4285	[-0.4479, -0.4085]
ViT-B/16	all	gt	attention_output_identity_tv	100	0.1823	[0.1692, 0.1956]
ViT-B/16	all	top1	attention_output_identity_tv	100	0.1848	[0.1723, 0.1978]
ViT-B/16	all	margin	attention_output_identity_tv	100	0.1233	[0.1128, 0.1340]
ViT-B/16	all	gt	attention_residual_path_cancellation	100	0.2017	[0.1688, 0.2320]
ViT-B/16	all	top1	attention_residual_path_cancellation	100	0.2042	[0.1719, 0.2345]
ViT-B/16	all	margin	attention_residual_path_cancellation	100	0.0329	[0.0108, 0.0553]

Model	Subset	Target	Metric	n	B11 - B0 mean	95% CI
ViT-B/16	all	gt	mlp__identity__scale	100	0.4057	[0.3937, 0.4183]
ViT-B/16	all	top1	mlp__identity__scale	100	0.4033	[0.3919, 0.4144]
ViT-B/16	all	margin	mlp__identity__scale	100	0.5437	[0.5254, 0.5627]
ViT-B/16	all	gt	mlp__output__identity__tv	100	-0.0275	[-0.0369, -0.0182]
ViT-B/16	all	top1	mlp__output__identity__tv	100	-0.0277	[-0.0370, -0.0186]
ViT-B/16	all	margin	mlp__output__identity__tv	100	-0.0897	[-0.0985, -0.0805]
ViT-B/16	all	gt	mlp__residual__path__cancellation	100	0.4451	[0.4210, 0.4662]
ViT-B/16	all	top1	mlp__residual__path__cancellation	100	0.4546	[0.4343, 0.4731]
ViT-B/16	all	margin	mlp__residual__path__cancellation	100	0.1074	[0.0777, 0.1369]
ViT-B/16	correct	gt	routing__scale	92	0.2058	[0.1891, 0.2222]
ViT-B/16	correct	top1	routing__scale	92	0.2058	[0.1891, 0.2222]
ViT-B/16	correct	margin	routing__scale	92	0.1467	[0.1299, 0.1633]
ViT-B/16	correct	gt	value__error	92	0.3623	[0.3256, 0.4001]
ViT-B/16	correct	top1	value__error	92	0.3623	[0.3256, 0.4001]
ViT-B/16	correct	margin	value__error	92	0.2813	[0.2476, 0.3116]
ViT-B/16	correct	gt	jvp__discrepancy	92	0.4829	[0.4657, 0.4998]
ViT-B/16	correct	top1	jvp__discrepancy	92	0.4829	[0.4657, 0.4998]
ViT-B/16	correct	margin	jvp__discrepancy	92	0.4829	[0.4657, 0.4998]
ViT-B/16	correct	gt	attention__path__cancellation	92	0.1605	[0.1480, 0.1723]
ViT-B/16	correct	top1	attention__path__cancellation	92	0.1605	[0.1480, 0.1723]
ViT-B/16	correct	margin	attention__path__cancellation	92	0.1453	[0.1339, 0.1561]
ViT-B/16	correct	gt	saturation__routing	92	-0.1206	[-0.2004, -0.0436]
ViT-B/16	correct	top1	saturation__routing	92	-0.1206	[-0.2004, -0.0436]
ViT-B/16	correct	margin	saturation__routing	92	-0.0131	[-0.0883, 0.0598]
ViT-B/16	correct	gt	saturation__error	92	-0.0628	[-0.1478, 0.0175]
ViT-B/16	correct	top1	saturation__error	92	-0.0628	[-0.1478, 0.0175]
ViT-B/16	correct	margin	saturation__error	92	0.0260	[-0.0562, 0.1081]
ViT-B/16	correct	gt	target__contracted__jvp	92	0.3431	[0.3039, 0.3823]
ViT-B/16	correct	top1	target__contracted__jvp	92	0.3431	[0.3039, 0.3823]
ViT-B/16	correct	margin	target__contracted__jvp	92	0.2953	[0.2618, 0.3283]
ViT-B/16	correct	gt	attention__identity__scale	92	-0.4362	[-0.4546, -0.4194]
ViT-B/16	correct	top1	attention__identity__scale	92	-0.4362	[-0.4546, -0.4194]
ViT-B/16	correct	margin	attention__identity__scale	92	-0.4281	[-0.4478, -0.4091]
ViT-B/16	correct	gt	attention__output__identity__tv	92	0.1884	[0.1759, 0.2016]
ViT-B/16	correct	top1	attention__output__identity__tv	92	0.1884	[0.1759, 0.2016]
ViT-B/16	correct	margin	attention__output__identity__tv	92	0.1259	[0.1143, 0.1374]
ViT-B/16	correct	gt	attention__residual__path__cancellation	92	0.2164	[0.1837, 0.2492]
ViT-B/16	correct	top1	attention__residual__path__cancellation	92	0.2164	[0.1837, 0.2492]
ViT-B/16	correct	margin	attention__residual__path__cancellation	92	0.0374	[0.0136, 0.0611]
ViT-B/16	correct	gt	mlp__identity__scale	92	0.4015	[0.3894, 0.4136]
ViT-B/16	correct	top1	mlp__identity__scale	92	0.4015	[0.3894, 0.4136]
ViT-B/16	correct	margin	mlp__identity__scale	92	0.5434	[0.5249, 0.5620]
ViT-B/16	correct	gt	mlp__output__identity__tv	92	-0.0285	[-0.0381, -0.0189]
ViT-B/16	correct	top1	mlp__output__identity__tv	92	-0.0285	[-0.0381, -0.0189]
ViT-B/16	correct	margin	mlp__output__identity__tv	92	-0.0900	[-0.0996, -0.0807]
ViT-B/16	correct	gt	mlp__residual__path__cancellation	92	0.4577	[0.4381, 0.4765]
ViT-B/16	correct	top1	mlp__residual__path__cancellation	92	0.4577	[0.4381, 0.4765]
ViT-B/16	correct	margin	mlp__residual__path__cancellation	92	0.1127	[0.0840, 0.1427]
ViT-B/16	incorrect	gt	routing__scale	8	0.1719	[0.1298, 0.2101]
ViT-B/16	incorrect	top1	routing__scale	8	0.1706	[0.1210, 0.2183]
ViT-B/16	incorrect	margin	routing__scale	8	0.1120	[0.0779, 0.1399]
ViT-B/16	incorrect	gt	value__error	8	0.2568	[0.1380, 0.3788]
ViT-B/16	incorrect	top1	value__error	8	0.2985	[0.1477, 0.4298]
ViT-B/16	incorrect	margin	value__error	8	0.2183	[0.1684, 0.2715]
ViT-B/16	incorrect	gt	jvp__discrepancy	8	0.3924	[0.2737, 0.5011]
ViT-B/16	incorrect	top1	jvp__discrepancy	8	0.3924	[0.2737, 0.5011]
ViT-B/16	incorrect	margin	jvp__discrepancy	8	0.3924	[0.2737, 0.5011]
ViT-B/16	incorrect	gt	attention__path__cancellation	8	0.1385	[0.1052, 0.1729]
ViT-B/16	incorrect	top1	attention__path__cancellation	8	0.1563	[0.1272, 0.1850]
ViT-B/16	incorrect	margin	attention__path__cancellation	8	0.1125	[0.0926, 0.1367]
ViT-B/16	incorrect	gt	saturation__routing	8	0.0411	[-0.2710, 0.3628]
ViT-B/16	incorrect	top1	saturation__routing	8	0.0271	[-0.2099, 0.2570]
ViT-B/16	incorrect	margin	saturation__routing	8	0.2780	[-0.0166, 0.5656]
ViT-B/16	incorrect	gt	saturation__error	8	0.1031	[-0.2203, 0.4493]
ViT-B/16	incorrect	top1	saturation__error	8	0.0760	[-0.1198, 0.2850]

Model	Subset	Target	Metric	n	B11 - B0 mean	95% CI
ViT-B/16	incorrect	margin	saturation_error	8	0.2273	[0.0402, 0.4187]
ViT-B/16	incorrect	gt	target_contracted_jvp	8	0.2814	[0.1617, 0.3988]
ViT-B/16	incorrect	top1	target_contracted_jvp	8	0.3210	[0.1766, 0.4598]
ViT-B/16	incorrect	margin	target_contracted_jvp	8	0.2126	[0.1223, 0.3130]
ViT-B/16	incorrect	gt	attention_identity_scale	8	-0.4712	[-0.5617, -0.3925]
ViT-B/16	incorrect	top1	attention_identity_scale	8	-0.4740	[-0.5316, -0.4160]
ViT-B/16	incorrect	margin	attention_identity_scale	8	-0.4324	[-0.5205, -0.3503]
ViT-B/16	incorrect	gt	attention_output_identity_tv	8	0.1119	[0.0867, 0.1380]
ViT-B/16	incorrect	top1	attention_output_identity_tv	8	0.1432	[0.1076, 0.1813]
ViT-B/16	incorrect	margin	attention_output_identity_tv	8	0.0931	[0.0694, 0.1182]
ViT-B/16	incorrect	gt	attention_residual_path_cancellation	8	0.0321	[-0.0432, 0.1290]
ViT-B/16	incorrect	top1	attention_residual_path_cancellation	8	0.0640	[-0.0001, 0.1464]
ViT-B/16	incorrect	margin	attention_residual_path_cancellation	8	-0.0194	[-0.0629, 0.0265]
ViT-B/16	incorrect	gt	mlp_identity_scale	8	0.4537	[0.4078, 0.5139]
ViT-B/16	incorrect	top1	mlp_identity_scale	8	0.4232	[0.3842, 0.4573]
ViT-B/16	incorrect	margin	mlp_identity_scale	8	0.5469	[0.4811, 0.6039]
ViT-B/16	incorrect	gt	mlp_output_identity_tv	8	-0.0169	[-0.0455, 0.0156]
ViT-B/16	incorrect	top1	mlp_output_identity_tv	8	-0.0190	[-0.0503, 0.0132]
ViT-B/16	incorrect	margin	mlp_output_identity_tv	8	-0.0862	[-0.1020, -0.0693]
ViT-B/16	incorrect	gt	mlp_residual_path_cancellation	8	0.3002	[0.1584, 0.4207]
ViT-B/16	incorrect	top1	mlp_residual_path_cancellation	8	0.4195	[0.3318, 0.5030]
ViT-B/16	incorrect	margin	mlp_residual_path_cancellation	8	0.0468	[-0.0459, 0.1747]
DeiT-S/16	all	gt	routing_scale	100	0.2617	[0.2450, 0.2785]
DeiT-S/16	all	top1	routing_scale	100	0.2607	[0.2438, 0.2778]
DeiT-S/16	all	margin	routing_scale	100	0.2460	[0.2302, 0.2625]
DeiT-S/16	all	gt	value_error	100	0.5806	[0.5452, 0.6180]
DeiT-S/16	all	top1	value_error	100	0.5745	[0.5354, 0.6131]
DeiT-S/16	all	margin	value_error	100	0.5744	[0.5352, 0.6135]
DeiT-S/16	all	gt	jvp_discrepancy	100	0.6374	[0.6193, 0.6563]
DeiT-S/16	all	top1	jvp_discrepancy	100	0.6374	[0.6193, 0.6563]
DeiT-S/16	all	margin	jvp_discrepancy	100	0.6374	[0.6193, 0.6563]
DeiT-S/16	all	gt	attention_path_cancellation	100	0.2407	[0.2279, 0.2540]
DeiT-S/16	all	top1	attention_path_cancellation	100	0.2392	[0.2263, 0.2529]
DeiT-S/16	all	margin	attention_path_cancellation	100	0.2318	[0.2209, 0.2430]
DeiT-S/16	all	gt	saturation_routing	100	-0.3766	[-0.4594, -0.2909]
DeiT-S/16	all	top1	saturation_routing	100	-0.3789	[-0.4634, -0.2920]
DeiT-S/16	all	margin	saturation_routing	100	-0.2943	[-0.3863, -0.2006]
DeiT-S/16	all	gt	saturation_error	100	-0.3371	[-0.4417, -0.2337]
DeiT-S/16	all	top1	saturation_error	100	-0.3486	[-0.4509, -0.2480]
DeiT-S/16	all	margin	saturation_error	100	-0.0817	[-0.1794, 0.0137]
DeiT-S/16	all	gt	target_contracted_jvp	100	0.5393	[0.4936, 0.5852]
DeiT-S/16	all	top1	target_contracted_jvp	100	0.5523	[0.5020, 0.6041]
DeiT-S/16	all	margin	target_contracted_jvp	100	0.5375	[0.4887, 0.5895]
DeiT-S/16	all	gt	attention_identity_scale	100	-0.2168	[-0.2369, -0.1960]
DeiT-S/16	all	top1	attention_identity_scale	100	-0.2216	[-0.2418, -0.2006]
DeiT-S/16	all	margin	attention_identity_scale	100	-0.1014	[-0.1224, -0.0796]
DeiT-S/16	all	gt	attention_output_identity_tv	100	0.0821	[0.0723, 0.0918]
DeiT-S/16	all	top1	attention_output_identity_tv	100	0.0859	[0.0759, 0.0961]
DeiT-S/16	all	margin	attention_output_identity_tv	100	0.0337	[0.0255, 0.0420]
DeiT-S/16	all	gt	attention_residual_path_cancellation	100	0.0709	[0.0464, 0.0949]
DeiT-S/16	all	top1	attention_residual_path_cancellation	100	0.0706	[0.0456, 0.0950]
DeiT-S/16	all	margin	attention_residual_path_cancellation	100	-0.0152	[-0.0397, 0.0084]
DeiT-S/16	all	gt	mlp_identity_scale	100	0.3029	[0.2846, 0.3227]
DeiT-S/16	all	top1	mlp_identity_scale	100	0.3041	[0.2847, 0.3235]
DeiT-S/16	all	margin	mlp_identity_scale	100	0.4663	[0.4484, 0.4830]
DeiT-S/16	all	gt	mlp_output_identity_tv	100	-0.0414	[-0.0564, -0.0263]
DeiT-S/16	all	top1	mlp_output_identity_tv	100	-0.0413	[-0.0566, -0.0252]
DeiT-S/16	all	margin	mlp_output_identity_tv	100	-0.1197	[-0.1312, -0.1073]
DeiT-S/16	all	gt	mlp_residual_path_cancellation	100	0.2912	[0.2535, 0.3278]
DeiT-S/16	all	top1	mlp_residual_path_cancellation	100	0.3101	[0.2762, 0.3429]
DeiT-S/16	all	margin	mlp_residual_path_cancellation	100	-0.0462	[-0.0768, -0.0155]
DeiT-S/16	correct	gt	routing_scale	86	0.2590	[0.2389, 0.2772]
DeiT-S/16	correct	top1	routing_scale	86	0.2590	[0.2389, 0.2772]
DeiT-S/16	correct	margin	routing_scale	86	0.2445	[0.2258, 0.2624]
DeiT-S/16	correct	gt	value_error	86	0.5713	[0.5310, 0.6118]

Model	Subset	Target	Metric	n	B11 - B0 mean	95% CI
DeiT-S/16	correct	top1	value_error	86	0.5713	[0.5310, 0.6118]
DeiT-S/16	correct	margin	value_error	86	0.5580	[0.5160, 0.5994]
DeiT-S/16	correct	gt	jvp_discrepancy	86	0.6362	[0.6153, 0.6562]
DeiT-S/16	correct	top1	jvp_discrepancy	86	0.6362	[0.6153, 0.6562]
DeiT-S/16	correct	margin	jvp_discrepancy	86	0.6362	[0.6153, 0.6562]
DeiT-S/16	correct	gt	attention_path_cancellation	86	0.2396	[0.2250, 0.2535]
DeiT-S/16	correct	top1	attention_path_cancellation	86	0.2396	[0.2250, 0.2535]
DeiT-S/16	correct	margin	attention_path_cancellation	86	0.2285	[0.2155, 0.2406]
DeiT-S/16	correct	gt	saturation_routing	86	-0.4027	[-0.4930, -0.3083]
DeiT-S/16	correct	top1	saturation_routing	86	-0.4027	[-0.4930, -0.3083]
DeiT-S/16	correct	margin	saturation_routing	86	-0.3116	[-0.4140, -0.2066]
DeiT-S/16	correct	gt	saturation_error	86	-0.3841	[-0.4930, -0.2638]
DeiT-S/16	correct	top1	saturation_error	86	-0.3841	[-0.4930, -0.2638]
DeiT-S/16	correct	margin	saturation_error	86	-0.0850	[-0.1894, 0.0159]
DeiT-S/16	correct	gt	target_contracted_jvp	86	0.5355	[0.4808, 0.5896]
DeiT-S/16	correct	top1	target_contracted_jvp	86	0.5355	[0.4808, 0.5896]
DeiT-S/16	correct	margin	target_contracted_jvp	86	0.5281	[0.4708, 0.5831]
DeiT-S/16	correct	gt	attention_identity_scale	86	-0.2298	[-0.2518, -0.2078]
DeiT-S/16	correct	top1	attention_identity_scale	86	-0.2298	[-0.2518, -0.2078]
DeiT-S/16	correct	margin	attention_identity_scale	86	-0.1054	[-0.1286, -0.0821]
DeiT-S/16	correct	gt	attention_output_identity_tv	86	0.0885	[0.0775, 0.0999]
DeiT-S/16	correct	top1	attention_output_identity_tv	86	0.0885	[0.0775, 0.0999]
DeiT-S/16	correct	margin	attention_output_identity_tv	86	0.0356	[0.0266, 0.0449]
DeiT-S/16	correct	gt	attention_residual_path_cancellation	86	0.0827	[0.0547, 0.1116]
DeiT-S/16	correct	top1	attention_residual_path_cancellation	86	0.0827	[0.0547, 0.1116]
DeiT-S/16	correct	margin	attention_residual_path_cancellation	86	-0.0110	[-0.0399, 0.0195]
DeiT-S/16	correct	gt	mlp_identity_scale	86	0.2928	[0.2735, 0.3141]
DeiT-S/16	correct	top1	mlp_identity_scale	86	0.2928	[0.2735, 0.3141]
DeiT-S/16	correct	margin	mlp_identity_scale	86	0.4574	[0.4379, 0.4762]
DeiT-S/16	correct	gt	mlp_output_identity_tv	86	-0.0346	[-0.0509, -0.0184]
DeiT-S/16	correct	top1	mlp_output_identity_tv	86	-0.0346	[-0.0509, -0.0184]
DeiT-S/16	correct	margin	mlp_output_identity_tv	86	-0.1145	[-0.1268, -0.1011]
DeiT-S/16	correct	gt	mlp_residual_path_cancellation	86	0.3265	[0.2927, 0.3609]
DeiT-S/16	correct	top1	mlp_residual_path_cancellation	86	0.3265	[0.2927, 0.3609]
DeiT-S/16	correct	margin	mlp_residual_path_cancellation	86	-0.0345	[-0.0692, 0.0001]
DeiT-S/16	incorrect	gt	routing_scale	14	0.2785	[0.2462, 0.3109]
DeiT-S/16	incorrect	top1	routing_scale	14	0.2717	[0.2366, 0.3090]
DeiT-S/16	incorrect	margin	routing_scale	14	0.2551	[0.2297, 0.2818]
DeiT-S/16	incorrect	gt	value_error	14	0.6374	[0.5562, 0.7293]
DeiT-S/16	incorrect	top1	value_error	14	0.5942	[0.4738, 0.7350]
DeiT-S/16	incorrect	margin	value_error	14	0.6752	[0.5912, 0.7607]
DeiT-S/16	incorrect	gt	jvp_discrepancy	14	0.6446	[0.6040, 0.6821]
DeiT-S/16	incorrect	top1	jvp_discrepancy	14	0.6446	[0.6040, 0.6821]
DeiT-S/16	incorrect	margin	jvp_discrepancy	14	0.6446	[0.6040, 0.6821]
DeiT-S/16	incorrect	gt	attention_path_cancellation	14	0.2471	[0.2123, 0.2797]
DeiT-S/16	incorrect	top1	attention_path_cancellation	14	0.2369	[0.1957, 0.2831]
DeiT-S/16	incorrect	margin	attention_path_cancellation	14	0.2521	[0.2257, 0.2803]
DeiT-S/16	incorrect	gt	saturation_routing	14	-0.2163	[-0.4408, 0.0163]
DeiT-S/16	incorrect	top1	saturation_routing	14	-0.2327	[-0.4122, -0.0448]
DeiT-S/16	incorrect	margin	saturation_routing	14	-0.1878	[-0.3878, 0.0286]
DeiT-S/16	incorrect	gt	saturation_error	14	-0.0490	[-0.3469, 0.2367]
DeiT-S/16	incorrect	top1	saturation_error	14	-0.1306	[-0.3633, 0.0939]
DeiT-S/16	incorrect	margin	saturation_error	14	-0.0612	[-0.3674, 0.2327]
DeiT-S/16	incorrect	gt	target_contracted_jvp	14	0.5631	[0.5059, 0.6286]
DeiT-S/16	incorrect	top1	target_contracted_jvp	14	0.6555	[0.4921, 0.8523]
DeiT-S/16	incorrect	margin	target_contracted_jvp	14	0.5956	[0.4935, 0.6992]
DeiT-S/16	incorrect	gt	attention_identity_scale	14	-0.1368	[-0.1879, -0.0917]
DeiT-S/16	incorrect	top1	attention_identity_scale	14	-0.1713	[-0.2423, -0.1122]
DeiT-S/16	incorrect	margin	attention_identity_scale	14	-0.0768	[-0.1418, -0.0212]
DeiT-S/16	incorrect	gt	attention_output_identity_tv	14	0.0426	[0.0273, 0.0589]
DeiT-S/16	incorrect	top1	attention_output_identity_tv	14	0.0700	[0.0394, 0.1080]
DeiT-S/16	incorrect	margin	attention_output_identity_tv	14	0.0219	[0.0026, 0.0449]
DeiT-S/16	incorrect	gt	attention_residual_path_cancellation	14	-0.0021	[-0.0379, 0.0375]
DeiT-S/16	incorrect	top1	attention_residual_path_cancellation	14	-0.0041	[-0.0474, 0.0455]
DeiT-S/16	incorrect	margin	attention_residual_path_cancellation	14	-0.0407	[-0.0834, 0.0012]

Model	Subset	Target	Metric	n	B11 - B0 mean	95% CI
DeiT-S/16	incorrect	gt	mlp__identity__scale	14	0.3651	[0.3177, 0.4134]
DeiT-S/16	incorrect	top1	mlp__identity__scale	14	0.3737	[0.3208, 0.4260]
DeiT-S/16	incorrect	margin	mlp__identity__scale	14	0.5210	[0.4896, 0.5517]
DeiT-S/16	incorrect	gt	mlp__output__identity__tv	14	-0.0829	[-0.1111, -0.0537]
DeiT-S/16	incorrect	top1	mlp__output__identity__tv	14	-0.0822	[-0.1229, -0.0378]
DeiT-S/16	incorrect	margin	mlp__output__identity__tv	14	-0.1514	[-0.1788, -0.1242]
DeiT-S/16	incorrect	gt	mlp__residual__path__cancellation	14	0.0745	[-0.0319, 0.1855]
DeiT-S/16	incorrect	top1	mlp__residual__path__cancellation	14	0.2091	[0.1068, 0.3061]
DeiT-S/16	incorrect	margin	mlp__residual__path__cancellation	14	-0.1185	[-0.1821, -0.0485]

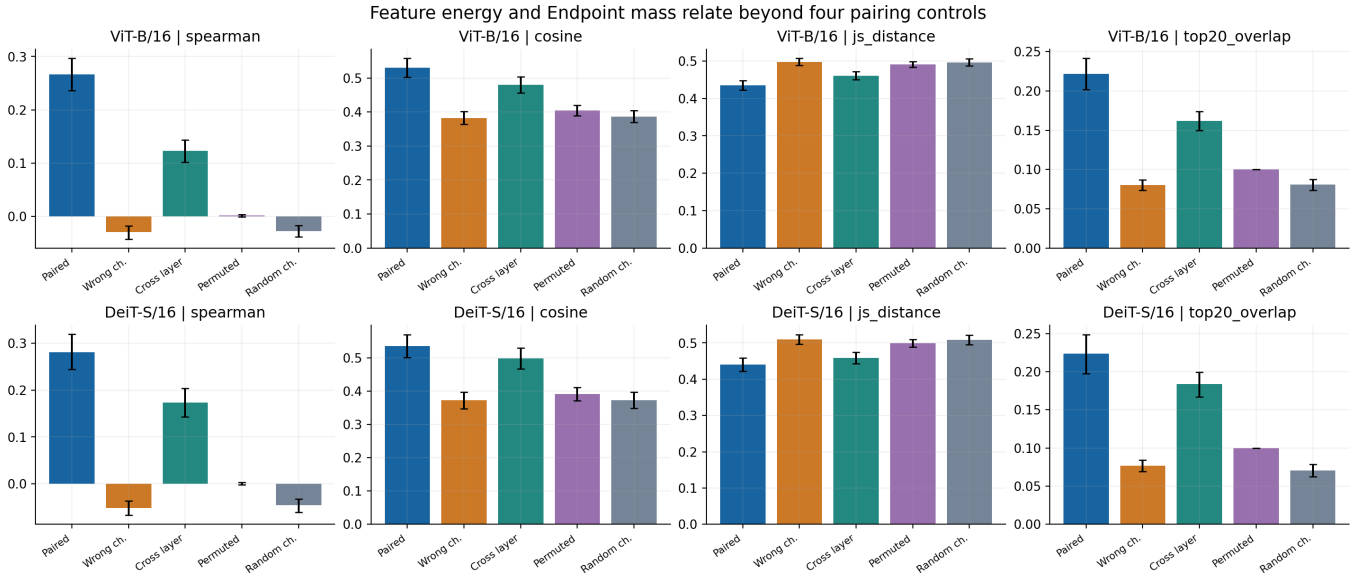
All block estimates, paired contrasts and NA counts · Image-block values

S12. Quantitative feature–Endpoint relationship

Each model uses 32 frozen images, 12 blocks and 16 frozen random channels per block, giving 12,288 energy targets. The sum covers 196 patch tokens and excludes CLS, matching the energy definition in the forward displays. Raw RGB Endpoint absolute mass is aggregated to patches; display normalization and spatial-variance channel selection do not enter the analysis.

$$s_{b,c} = \frac{1}{2}\sum_{t=1}^{196} h_{b,t,c}^2, \qquad q_t = \frac{h_{b,t,c}^2}{\sum_{j=1}^{196} h_{b,j,c}^2}, \qquad p_t = \frac{\sum_{i\in P_t} |x_i \partial s_{b,c}/\partial x_i|}{\sum_i |x_i \partial s_{b,c}/\partial x_i|}. \tag{2}$$

Mean image-level Spearman is 0.2663/0.2811 for correct pairing, with positive paired-difference intervals against wrong-channel, cross-layer same-channel, patch-permutation and random-channel controls. Cosine, Jensen–Shannon distance and top-20 overlap are also reported. Same-channel forward energy and local input dependence are associated beyond these controls, with substantial departures from equivalence. Channel and block medians are taken within each image before image-level statistics.



Model	Pairing	Metric	Mean	95% CI
ViT-B/16	paired	spearman	0.2663	[0.2366, 0.2973]

Model	Pairing	Metric	Mean	95% CI
ViT-B/16	wrong_channel	spearman	-0.0298	[-0.0426, -0.0177]
ViT-B/16	cross_layer	spearman	0.1232	[0.1019, 0.1435]
ViT-B/16	permuted	spearman	0.0013	[-0.0010, 0.0036]
ViT-B/16	random_channel	spearman	-0.0275	[-0.0389, -0.0175]
ViT-B/16	paired	cosine	0.5304	[0.5029, 0.5575]
ViT-B/16	wrong_channel	cosine	0.3826	[0.3643, 0.4011]
ViT-B/16	cross_layer	cosine	0.4802	[0.4562, 0.5032]
ViT-B/16	permuted	cosine	0.4043	[0.3888, 0.4198]
ViT-B/16	random_channel	cosine	0.3868	[0.3691, 0.4041]
ViT-B/16	paired	js_distance	0.4351	[0.4223, 0.4484]
ViT-B/16	wrong_channel	js_distance	0.4974	[0.4884, 0.5070]
ViT-B/16	cross_layer	js_distance	0.4604	[0.4498, 0.4716]
ViT-B/16	permuted	js_distance	0.4909	[0.4837, 0.4983]
ViT-B/16	random_channel	js_distance	0.4960	[0.4869, 0.5059]
ViT-B/16	paired	top20_overlap	0.2215	[0.2016, 0.2414]
ViT-B/16	wrong_channel	top20_overlap	0.0805	[0.0734, 0.0871]
ViT-B/16	cross_layer	top20_overlap	0.1621	[0.1496, 0.1742]
ViT-B/16	permuted	top20_overlap	0.1000	[0.1000, 0.1000]
ViT-B/16	random_channel	top20_overlap	0.0809	[0.0738, 0.0875]
DeiT-S/16	paired	spearman	0.2811	[0.2442, 0.3191]
DeiT-S/16	wrong_channel	spearman	-0.0514	[-0.0670, -0.0361]
DeiT-S/16	cross_layer	spearman	0.1735	[0.1430, 0.2036]
DeiT-S/16	permuted	spearman	0.0003	[-0.0021, 0.0029]
DeiT-S/16	random_channel	spearman	-0.0458	[-0.0611, -0.0318]
DeiT-S/16	paired	cosine	0.5359	[0.5010, 0.5697]
DeiT-S/16	wrong_channel	cosine	0.3723	[0.3477, 0.3966]
DeiT-S/16	cross_layer	cosine	0.4989	[0.4673, 0.5300]
DeiT-S/16	permuted	cosine	0.3912	[0.3718, 0.4111]
DeiT-S/16	random_channel	cosine	0.3733	[0.3490, 0.3974]
DeiT-S/16	paired	js_distance	0.4406	[0.4225, 0.4587]
DeiT-S/16	wrong_channel	js_distance	0.5094	[0.4962, 0.5228]
DeiT-S/16	cross_layer	js_distance	0.4584	[0.4429, 0.4740]
DeiT-S/16	permuted	js_distance	0.4990	[0.4886, 0.5098]
DeiT-S/16	random_channel	js_distance	0.5085	[0.4958, 0.5218]
DeiT-S/16	paired	top20_overlap	0.2234	[0.1973, 0.2484]
DeiT-S/16	wrong_channel	top20_overlap	0.0770	[0.0695, 0.0844]
DeiT-S/16	cross_layer	top20_overlap	0.1836	[0.1668, 0.1992]
DeiT-S/16	permuted	top20_overlap	0.1000	[0.1000, 0.1000]
DeiT-S/16	random_channel	top20_overlap	0.0707	[0.0625, 0.0785]

Model	Control	Metric	Paired similarity advantage	95% CI
ViT-B/16	wrong_channel	spearman	0.2961	[0.2763, 0.3165]
ViT-B/16	cross_layer	spearman	0.1431	[0.1292, 0.1579]
ViT-B/16	permuted	spearman	0.2649	[0.2350, 0.2961]
ViT-B/16	random_channel	spearman	0.2937	[0.2709, 0.3183]
ViT-B/16	wrong_channel	cosine	0.1478	[0.1367, 0.1584]
ViT-B/16	cross_layer	cosine	0.0502	[0.0429, 0.0582]
ViT-B/16	permuted	cosine	0.1261	[0.1126, 0.1391]
ViT-B/16	random_channel	cosine	0.1436	[0.1318, 0.1543]
ViT-B/16	wrong_channel	js_distance	0.0623	[0.0579, 0.0666]
ViT-B/16	cross_layer	js_distance	0.0253	[0.0220, 0.0286]
ViT-B/16	permuted	js_distance	0.0558	[0.0495, 0.0617]
ViT-B/16	random_channel	js_distance	0.0608	[0.0562, 0.0654]
ViT-B/16	wrong_channel	top20_overlap	0.1410	[0.1262, 0.1570]
ViT-B/16	cross_layer	top20_overlap	0.0594	[0.0484, 0.0711]
ViT-B/16	permuted	top20_overlap	0.1215	[0.1016, 0.1414]
ViT-B/16	random_channel	top20_overlap	0.1406	[0.1246, 0.1574]
DeiT-S/16	wrong_channel	spearman	0.3325	[0.3092, 0.3568]
DeiT-S/16	cross_layer	spearman	0.1076	[0.0975, 0.1181]
DeiT-S/16	permuted	spearman	0.2807	[0.2430, 0.3192]
DeiT-S/16	random_channel	spearman	0.3268	[0.3009, 0.3532]
DeiT-S/16	wrong_channel	cosine	0.1636	[0.1527, 0.1741]

Model	Control	Metric	Paired similarity advantage	95% CI
DeiT-S/16	cross_layer	cosine	0.0370	[0.0315, 0.0425]
DeiT-S/16	permuted	cosine	0.1447	[0.1259, 0.1633]
DeiT-S/16	random_channel	cosine	0.1626	[0.1507, 0.1740]
DeiT-S/16	wrong_channel	js_distance	0.0687	[0.0638, 0.0737]
DeiT-S/16	cross_layer	js_distance	0.0177	[0.0146, 0.0207]
DeiT-S/16	permuted	js_distance	0.0584	[0.0501, 0.0663]
DeiT-S/16	random_channel	js_distance	0.0679	[0.0628, 0.0730]
DeiT-S/16	wrong_channel	top20_overlap	0.1465	[0.1262, 0.1660]
DeiT-S/16	cross_layer	top20_overlap	0.0398	[0.0297, 0.0504]
DeiT-S/16	permuted	top20_overlap	0.1234	[0.0973, 0.1484]
DeiT-S/16	random_channel	top20_overlap	0.1527	[0.1328, 0.1723]

Unit metrics · Image-block values

S13. Sampling sensitivity

The same 32 images reuse all original M3 32 channels and 15 patch strata. Channel and token-stratum bootstraps use 256 draws each, image bootstrap uses 2,000 draws, and each block or spatial stratum is omitted in turn. The table gives resampling and deletion-sensitivity ranges. The three spatial trend directions persist across all ranges. This resampling analysis does not constitute independent channel-seed replication. CLS-minus-patch is the all-layer difference; other metrics are late-minus-early.

Model	Metric	Reference mean	Image CI	Channel bootstrap range	Token bootstrap range	Leave-one-block range	Leave-one-stratum range
ViT-B/16	entropy	0.2392	[0.2303, 0.2480]	[0.2307, 0.2472]	[0.2326, 0.2421]	[0.1847, 0.2937]	[0.2375, 0.2403]
ViT-B/16	locality	-0.3607	[-0.3683, -0.3526]	[-0.3678, -0.3523]	[-0.3641, -0.3526]	[-0.4091, -0.3123]	[-0.3617, -0.3596]
ViT-B/16	distance	0.1331	[0.1274, 0.1389]	[0.1295, 0.1358]	[0.1168, 0.1478]	[0.1182, 0.1479]	[0.1296, 0.1366]
ViT-B/16	cancellation	0.0024	[0.0021, 0.0028]	[0.0020, 0.0029]	[0.0022, 0.0027]	[0.0016, 0.0032]	[0.0024, 0.0025]
ViT-B/16	cls_minus_patch_entropy	0.0564	[0.0454, 0.0669]	[0.0551, 0.0590]	[0.0527, 0.0636]	[0.0528, 0.0600]	[0.0548, 0.0580]
DeiT-S/16	entropy	0.2236	[0.2167, 0.2299]	[0.2190, 0.2279]	[0.2185, 0.2274]	[0.1800, 0.2673]	[0.2232, 0.2254]
DeiT-S/16	locality	-0.3412	[-0.3527, -0.3300]	[-0.3447, -0.3376]	[-0.3494, -0.3315]	[-0.3797, -0.3026]	[-0.3429, -0.3378]
DeiT-S/16	distance	0.1261	[0.1191, 0.1338]	[0.1249, 0.1277]	[0.1138, 0.1381]	[0.1176, 0.1347]	[0.1239, 0.1304]
DeiT-S/16	cancellation	0.0020	[0.0016, 0.0025]	[0.0018, 0.0022]	[0.0018, 0.0024]	[0.0014, 0.0025]	[0.0019, 0.0021]
DeiT-S/16	cls_minus_patch_entropy	0.0829	[0.0699, 0.0943]	[0.0813, 0.0850]	[0.0741, 0.0931]	[0.0749, 0.0909]	[0.0794, 0.0848]

S14. Cross-layer all-head M2 and scope

The main block-5 three-head experiment corresponds to the original confirmatory protocol. Sealed M2 covers blocks 2/5/10, two models, all heads, two references and 400 images per family. Under native patch geometry, 11/12 full-input and 12/12 isolated-head families favor regional Taylor. The table retains Holm correction over the original 672 bidirectional AUC tests.

Model	Reference	Block	Outcome	n images	Taylor > Endpoint Holm p	Supported
DeiT-S/16	blur	10	full	400	5.841e-20	True
DeiT-S/16	blur	10	isolated	400	2.355e-82	True
DeiT-S/16	normalized-zero	10	full	400	0.0282	True
DeiT-S/16	normalized-zero	10	isolated	400	4.045e-90	True
DeiT-S/16	blur	2	full	400	1.103e-08	True
DeiT-S/16	blur	2	isolated	400	6.597e-69	True

Model	Reference	Block	Outcome	n images	Taylor > Endpoint Holm p	Supported
DeiT-S/16	normalized-zero	2	full	400	0.04217	True
DeiT-S/16	normalized-zero	2	isolated	400	2.276e-58	True
DeiT-S/16	blur	5	full	400	6.793e-15	True
DeiT-S/16	blur	5	isolated	400	7.52e-85	True
DeiT-S/16	normalized-zero	5	full	400	0.5283	False
DeiT-S/16	normalized-zero	5	isolated	400	2.355e-82	True
ViT-B/16	blur	10	full	400	5.562e-28	True
ViT-B/16	blur	10	isolated	400	2.351e-94	True
ViT-B/16	normalized-zero	10	full	400	2.309e-26	True
ViT-B/16	normalized-zero	10	isolated	400	2.424e-97	True
ViT-B/16	blur	2	full	400	4.341e-16	True
ViT-B/16	blur	2	isolated	400	6.185e-64	True
ViT-B/16	normalized-zero	2	full	400	2.49e-17	True
ViT-B/16	normalized-zero	2	isolated	400	7.981e-47	True
ViT-B/16	blur	5	full	400	2.591e-25	True
ViT-B/16	blur	5	isolated	400	1.679e-91	True
ViT-B/16	normalized-zero	5	full	400	9.405e-18	True
ViT-B/16	normalized-zero	5	isolated	400	1.042e-115	True

Optional cascading randomization, the P2 same-target M3–M4 bridge, independent channel seeds and alternative checkpoints, and P3 input transforms and CNN display were not executed. Training controls concern local parameter-dependent structure; different scalar targets do not form a unit-matched finite causal mechanism. Original Path-IG calibration failure and confirmatory decisions are retained.

Frozen protocol · Source bindings · Task acquisition · Numerical validation

S15. Data, models, populations, and analysis identities

ImageNet data come from the official ILSVRC2012 validation set. Models instantiate timm 0.9.16 standard Block/Attention and strictly load checkpoint keys. All checkpoint byte counts and SHA-256 values were recomputed. AugReg names identify IN21K pretraining followed by IN1K fine-tuning or IN1K training; DeiT-S uses public Facebook IN1K weights. Results condition on these parameter states.

Model ID	Exact timm name / training	Bytes	SHA-256
vit_b16	vit_base_patch16_224.augreg2_in21k	346329431	489bfa2f1aa3b6781e268a87eb629816b89eebba43942
deit_s16	deit_small_patch16_224.fb_in1k	88251143	cd65a15597004d0ce19d7a9daef969903972db5b398e3
vit_b16_in1k	vit_base_patch16_224.augreg_in1k	346335542	69c9a2822a47838f57e522350b10ccc0d89d67fdcee076
vit_s16_augreg	vit_small_patch16_224.augreg_in21k	88267574	545815b4e770d2fa6ca4b3ccba7c16b035e474354e52d

Discovery, calibration, and evaluation contain 64, 64, and 400 classes, respectively, with one image per class. The 100-class unit study is an evaluation subset. E1 has five images from each of 100 classes; E5 has one from each of 400 classes. E5 uses class_e5/image_e5 hash ordering after excluding historical exposures and E1. The matrix reports image identity overlap. Classes may recur across later cohorts; independent images do not imply independent classes. E5 train/validation/test labels belong to a separate prediction analysis and do not change this all-400-image H1–H4 check.

Cohort	discovery	calibration	evaluation	unit_study	e1	e5
discovery	64	0	0	0	0	0
calibration	0	64	0	0	0	0
evaluation	0	0	400	100	0	0
unit_study	0	0	100	100	0	0
e1	0	0	0	0	500	0

Cohort	discovery	calibration	evaluation	unit_study	e1	e5
e5	0	0	0	0	0	400

Preprocessing uses Pillow 10.2.0 RGB without EXIF transpose. Torchvision 0.17.2 resizes the PIL image's shorter side to 248 with bicubic interpolation and antialias=True, center-crops to 224, converts to float32, and divides by 255. Table 2 specifies model normalization. Blur operates after cropping in RGB with a 31×31 Gaussian kernel, sigma=10, and reflect padding=15, then normalizes. Evaluation mode disables dropout and drop path. CLS, positions, and biases are fixed. The original logit target uses the official ground-truth class; clean-top1 is a separate frozen readout. Neither determines cohort selection. The original registry records its four-token/four-channel per-block selections. M3 fixes CLS plus 15 tokens from 3×5 spatial strata and 32 channels per block. Equal cross-model channel indices do not imply semantic correspondence. Structural analysis scans all heads and 12 blocks; the primary regional task fixes block 5 with ViT-B heads 0/6/11 and DeiT-S heads 0/3/5.

Selections use deterministic rules or explicit seeds. Actual IDs and targets appear in cohort bindings. Original protocols retain exact random-control keys and invocation order; seeds from different analyses are not interchangeable.

Purpose	Frozen seed / deterministic rule
Original image selection	SHA256(NNS_VIT_ENDPOINT_EP0_R1_20260918 | class | WNID), then namespace | image | filename
Unit cohort	SHA256(ENDPOINT_M3_COHORT_20260919 | sample_id | image_sha256), lowest 100
Unit channels	SHA256(ENDPOINT_M3_CHANNEL_20260919 | model | block | channel), lowest 32
Unit tokens	CLS + nearest patch to each of 3×5 strata centers; row-major ties
Original bootstrap	SHA256(namespace | bootstrap | family_id), first 8 bytes big-endian; PCG64
Unit bootstrap	SHA256(ENDPOINT_M3_BOOTSTRAP_20260919 | M3 | model | metric), first 8 bytes
Structural bootstrap	SHA256(M4 | test | model | test_id), with residual type inserted where applicable
Random initialization	202609201, 202609202, 202609203
Independent cohort / target selection	NNS_VIT_REVISION_V5_20260921_R1 namespace; class_e5, image_e5 and post/block/channel/token keys
Independent-cohort bootstrap	ENDPOINT_M1_20260918_V1 | 20260918 | bootstrap | V5 | (model,hypothesis)
Coordinate/threshold bootstrap	SHA256(2026092302 | analysis key), first 8 bytes; 10,000 PCG64 replicates
Analytic directions	NumPy default_rng(2026092301), three normalized Gaussian directions
Regional random orders	Frozen namespace/model/image/reference/head/repeat keys in original intervention registry; 20 repeats

Confirmatory identity is relative to each locally frozen pre-analysis plan; no external registration is supplied. Families retain their original members and uses, without retrospective combination with the original 28 tests. The analysis identity record contains all 28 test IDs and source hashes. The 0.10/0.05 thresholds lack external utility calibration and are researcher-set operational choices. Original decisions jointly require validity, attempted coverage, n and interval precision, the point estimate, and Holm significance; CI lower > 0.10 is not required. All 68 threshold sensitivity rows are available.

Analysis	Identity	Multiplicity family	Local freeze evidence
original_component_and_region	confirmatory	28 tests jointly Holm; 12 component correlations + 16 regional baseline contrasts	local pre-outcome plan; first committed occurrence below; 6237ef87a1af0205973b76bee172defe33a0a6da 2026-09-18T09:26:20+08:00

Analysis	Identity	Multiplicity family	Local freeze evidence
unit_depth	confirmatory	five per model: entropy, locality, distance, cancellation, CLS-patch entropy	local pre-outcome plan; first committed occurrence below; f2eba3f60d765e6a9729d9ff12a57bd01f93f952 2026-09-19T13:53:48+08:00
attention_residual_depth	confirmatory	four attention tests/model; two residual tests/model/type	local pre-outcome plan; first committed occurrence below; 1a1000bcdd15d8b0e76406f4376a82d60ec79b86 2026-09-19T17:25:17+08:00
head_baselines	exploratory	separate paired and absolute Spearman families; other metrics descriptive	local pre-outcome plan; first committed occurrence below; 33c8c320446a631b2e451016b3268eefdd9c926b 2026-09-19T04:20:36+08:00
residual_spatial_tv	post-hoc	four Holm contrasts across model and residual type	2026-09-19T10:59:17.569991+00:00; protocol timestamp and hashes
QKV_spatial	prospective robustness	30 descriptive contrasts; no confirmatory family combined with prior tests	2026-09-20T17:14:57.250515+00:00; protocol timestamp and hashes
independent_cohort_checkpoints	prospective robustness	H1-H4 Holm separately within each checkpoint	local pre-outcome plan; first committed occurrence below; 944f54c99c8d83803d461224565b93a779e75915 2026-09-21T11:31:10+08:00
training_target_sampling_controls	exploratory	descriptive 2000 bootstrap; training control resamples image and initialization seed; sampling control resamples target strata and images	2026-09-20T00:22:59.620440+00:00; 395d30b3804983d0e51e99a28d0c3c900aca8fda 2026-09-20T22:41:48+08:00
coordinate_sensitivity_v6	prospective robustness	no significance-based success criterion; report all directions and intervals	2026-09-22T16:43:58.690139+00:00; protocol timestamp and hashes
threshold_sensitivity_v6	post-hoc	descriptive thresholds only; original tests and decisions immutable	2026-09-22T16:43:58.690139+00:00; protocol timestamp and hashes

S16. Finite-replacement variables and executable pseudocode

A and V are forward variables; routing-induced/value-induced are differential partitions. Table 6's A/V family contains one all-head A and one all-head V replacement per block, totaling 24 components. Head families include all block/head pairs; residual families contain attention and MLP branches per block. Reference activations come from a complete reference forward. The consumer slot is part of replacement identity.

Replacement	Forward tensor / shape (batch=1)	Consumer input slot	Clean state and recomputation
Attention probability	A [1,H,197,197], after softmax/dropout	left input of A @ V	V, other branches and residual identity clean; recompute AV, concat, output projection, residual suffix
V projection	V [1,H,197,64], qkv projection slice	right input of A @ V	Q/K/A and identity clean; recompute AV and entire suffix
Head output	slice of projection_input [1,197,H×64], head h channels h×64:(h+1)×64	attn.proj input channels	all other head slices and residual identity clean; recompute projection and suffix
Attention branch	drop_path1 output [1,197,D]	residual1 add branch input	identity u clean; recompute residual sum, MLP and remaining blocks
MLP branch	drop_path2 output [1,197,D]	residual2 add branch input	carry r1 clean; recompute residual sum and remaining blocks

```
clean_rgb = RGB(image) / 255
clean_rgb = center_crop(resize_shorter_side(clean_rgb, 248, bicubic, antialias=True), 224)
x = (clean_rgb - mu) / sigma
reference_zero = zeros_like(x)              # raw RGB equals mu
```

```
reference_blur = (GaussianBlur(clean_rgb, kernel=31, sigma=10, reflect_padding=15) - mu) / sigma
clean = forward_and_capture(x)
reference = forward_and_capture(reference_input)
for fixed block, component, consumer slot:
    replacement = reference.tensor_at(component, slot)
    patched = forward(x, replace_only_this_use=replacement)
    effect = clean.GT_logit - patched.GT_logit
    # All sibling heads and bypass inputs remain clean before the replacement.
    # Every node downstream of the replaced use is recomputed normally.

for singleton region P in the fixed 4x4 grid of 56x56 pixels:
    x_P = x.copy(); x_P[P] = reference_input[P]
    mixed[P] = forward_and_capture(x_P)
    regional_Taylor[P] = abs(dot(clean_head_cotangent, clean_head - mixed[P].head))
rank_regions(score, ties=ascending_region_id)
for budget in [1, 2, 4]:
    selected = top_regions(budget)
    x_selected = x with selected pixels from reference_input
    mixed = forward_and_capture(x_selected)
    full_input_effect = clean.GT_logit - mixed.GT_logit
    isolated = forward(x, target_head_slice=mixed.head_slice)
    isolated_effect = clean.GT_logit - isolated.GT_logit
AUC = trapezoid(abs(isolated_effect), areas=[0,1/16,2/16,4/16]) / 0.25
image_AUC = mean(AUC across the three fixed heads)
random_AUC = mean(AUC per random repeat, then heads) # abs before averaging repeats
```

Full-input effects change the entire forward computation. Isolated-component effects insert only the target tensor from a mixed-input forward into a clean forward, preserving sibling heads and bypass inputs. Candidate sets match across regional scores, but Taylor requires singleton mixed forwards and therefore additional information. Region ties use ascending IDs; head ties use ascending component IDs; Spearman uses average ranks. Candidate-envelope AUC takes the greatest evaluated absolute effect separately at each area, then integrates and averages heads. It may join points from different rankings and is not an executable globally optimal region policy. It supplies an upper envelope over evaluated candidates and the regret denominator.

S17. Coordinate sensitivity and independent analytic checks

All representations use the same current input point, fixed target, and checkpoint. Raw gradient divides by model sigma; the common-origin weighted gradient uses raw RGB zero, independently of model mean. Cells report image-level median differences and 95% bootstrap intervals. Unit n=100 and structural n=400 per model, with no missing images. Coordinates are paired within images and are not additional samples.

$$E_i^{(x)} = x_i \frac{\partial s}{\partial x_i}, \qquad G_i^{(r)} = \frac{1}{\sigma_i} \frac{\partial s}{\partial x_i}, \qquad E_i^{(0)} = r_i \frac{\partial s}{\partial r_i} = \left(x_i + \frac{\mu_i}{\sigma_i} \right) \frac{\partial s}{\partial x_i}. \tag{3}$$

Model	Object / metric	Contrast	Normalized Endpoint	Raw-pixel gradient	Common-origin gradient	n / NA
deit_s16	K / entropy	block11_minus_block0	-0.067 [-0.074, -0.059]	-0.066 [-0.073, -0.060]	-0.061 [-0.069, -0.053]	400 / 0
deit_s16	K / entropy	late_minus_early	-0.028 [-0.031, -0.024]	-0.029 [-0.032, -0.026]	-0.027 [-0.031, -0.024]	400 / 0
deit_s16	Q / entropy	block11_minus_block0	-0.004 [-0.008, 0.0005]	0.003 [-0.001, 0.006]	0.002 [-0.003, 0.006]	400 / 0
deit_s16	Q / entropy	late_minus_early	0.018 [0.013, 0.022]	0.018 [0.015, 0.022]	0.015 [0.012, 0.018]	400 / 0
deit_s16	V / entropy	block11_minus_block0	-0.067 [-0.075, -0.061]	-0.076 [-0.081, -0.068]	-0.070 [-0.077, -0.062]	400 / 0
deit_s16	V / entropy	late_minus_early	-0.013 [-0.016, -0.011]	-0.016 [-0.017, -0.013]	-0.015 [-0.017, -0.013]	400 / 0
deit_s16	attention / output_identity_tv	block11_minus_block0	0.078 [0.072, 0.085]	0.079 [0.070, 0.085]	0.074 [0.067, 0.081]	400 / 0
deit_s16	attention / output_identity_tv	late_minus_early	0.028 [0.026, 0.030]	0.028 [0.026, 0.030]	0.027 [0.026, 0.029]	400 / 0
deit_s16	mlp / output_identity_tv	block11_minus_block0	-0.050 [-0.059, -0.040]	-0.044 [-0.057, -0.037]	-0.037 [-0.043, -0.029]	400 / 0
deit_s16	mlp / output_identity_tv	late_minus_early	-0.046 [-0.052, -0.041]	-0.044 [-0.051, -0.039]	-0.039 [-0.044, -0.034]	400 / 0
deit_s16	unit / distance	block11_minus_block0	0.163 [0.159, 0.169]	0.169 [0.164, 0.175]	0.171 [0.167, 0.178]	100 / 0
deit_s16	unit / distance	late_minus_early	0.122 [0.120, 0.128]	0.129 [0.126, 0.132]	0.127 [0.122, 0.131]	100 / 0
deit_s16	unit / entropy	block11_minus_block0	0.397 [0.387, 0.404]	0.426 [0.412, 0.438]	0.418 [0.400, 0.427]	100 / 0
deit_s16	unit / entropy	late_minus_early	0.225 [0.212, 0.228]	0.245 [0.241, 0.249]	0.221 [0.217, 0.229]	100 / 0
deit_s16	unit / locality	block11_minus_block0	-0.455 [-0.467, -0.432]	-0.453 [-0.469, -0.441]	-0.477 [-0.493, -0.455]	100 / 0
deit_s16	unit / locality	late_minus_early	-0.342 [-0.353, -0.333]	-0.359 [-0.366, -0.352]	-0.356 [-0.361, -0.344]	100 / 0
vit_b16	K / entropy	block11_minus_block0	-0.059 [-0.064, -0.053]	-0.052 [-0.058, -0.048]	-0.046 [-0.054, -0.041]	400 / 0
vit_b16	K / entropy	late_minus_early	-0.021 [-0.024, -0.019]	-0.019 [-0.023, -0.016]	-0.019 [-0.022, -0.016]	400 / 0
vit_b16	Q / entropy	block11_minus_block0	0.027 [0.022, 0.031]	0.026 [0.018, 0.031]	0.014 [0.009, 0.018]	400 / 0
vit_b16	Q / entropy	late_minus_early	0.037 [0.033, 0.043]	0.039 [0.033, 0.044]	0.034 [0.030, 0.037]	400 / 0
vit_b16	V / entropy	block11_minus_block0	-0.053 [-0.059, -0.047]	-0.057 [-0.061, -0.049]	-0.055 [-0.060, -0.050]	400 / 0
vit_b16	V / entropy	late_minus_early	-0.010 [-0.012, -0.008]	-0.009 [-0.011, -0.007]	-0.010 [-0.012, -0.007]	400 / 0
vit_b16	attention / output_identity_tv	block11_minus_block0	0.166 [0.159, 0.172]	0.167 [0.159, 0.174]	0.167 [0.160, 0.175]	400 / 0
vit_b16	attention / output_identity_tv	late_minus_early	0.052 [0.049, 0.055]	0.054 [0.051, 0.056]	0.053 [0.050, 0.055]	400 / 0
vit_b16	mlp / output_identity_tv	block11_minus_block0	-0.037 [-0.043, -0.031]	-0.034 [-0.038, -0.029]	-0.032 [-0.039, -0.025]	400 / 0
vit_b16	mlp / output_identity_tv	late_minus_early	-0.019 [-0.022, -0.015]	-0.017 [-0.020, -0.013]	-0.018 [-0.021, -0.014]	400 / 0
vit_b16	unit / distance	block11_minus_block0	0.197 [0.190, 0.204]	0.205 [0.200, 0.211]	0.209 [0.203, 0.214]	100 / 0
vit_b16	unit / distance	late_minus_early	0.131 [0.129, 0.136]	0.138 [0.134, 0.141]	0.134 [0.129, 0.138]	100 / 0
vit_b16	unit / entropy	block11_minus_block0	0.398 [0.392, 0.409]	0.440 [0.426, 0.447]	0.412 [0.402, 0.426]	100 / 0
vit_b16	unit / entropy	late_minus_early	0.220 [0.211, 0.227]	0.240 [0.235, 0.246]	0.217 [0.209, 0.223]	100 / 0
vit_b16	unit / locality	block11_minus_block0	-0.526 [-0.534, -0.516]	-0.543 [-0.554, -0.532]	-0.545 [-0.552, -0.538]	100 / 0
vit_b16	unit / locality	late_minus_early	-0.356 [-0.363, -0.350]	-0.367 [-0.373, -0.361]	-0.356 [-0.364, -0.347]	100 / 0

DeiT-S Q changes B11−B0 point-estimate sign across coordinates and all three intervals cross zero. Differences between phase and endpoint contrasts also show why an endpoint comparison cannot replace a trajectory. Main claims remain restricted to normalized-input Endpoint. Other objects support direction retention for these three representations, not general parameterization invariance.

The independent NumPy float64 reference explicitly constructs affine, LayerNorm, softmax $\mathrm{diag}(a) - aa^{\mathsf{T}}$, Q/K/V,

head concatenation/output projection, GELU/MLP, and residual-identity Jacobians. It imports no scanner, hooks, or autograd. A separate driver compares reference maps with the actual main scanner on single-head, double-head, and pre-LN residual attention/MLP tiny models. All 32 component comparisons and nine frozen-direction finite differences pass, with maximum component absolute error 1.11×10^{-16}. Per-component relative L2 errors and tolerances appear in analytic results. Shared-producer wrong-slot, attention Q/K-slot exchange, premature fan-out merging, and duplicated merged terms are rejected after the correct implementation passes. Coverage is limited to these numerical graphs and fault types, rather than all possible scanner errors.

The legacy regional key raw_gradient stores $\nabla_x s$, not $\nabla_r s$; the main text names its actual coordinates. Random region order hashes the joined NNS_VIT_ENDPOINT_E5_RANDOM_REGIONS_V1, task_seed, model_id, reference_id, head, and repeat string with SHA-256, uses its first eight big-endian bytes as the PCG64 seed, then applies permutation(16). task_seed derives from NNS_VIT_ENDPOINT_QUEUE_V1 and task_id.

S18. Revision-v7 type, depth, and quality diagnostics

This section uses the active CPU-only nns_vit_endpoint_revision_v7_r4_20260923 root. R1–R3 retain their statistical or provenance failures. R4 restores the R2 bootstrap seed, keeps concordance descriptive, aggregates residual-TV quality strata within images, and binds the final analysis, validator, and test hashes before execution. Historical pooled families, the 28-test Holm family, and all sealed decisions remain unchanged.

Table S18.1. Type-specific correlation; image-level median [95% CI]. Each value is computed across 12 blocks within an image and is a post-hoc diagnostic.

Model	Reference	A only	V only	Attention branch only	MLP branch only
vit_b16	normalized-zero	0.049 [0.007, 0.091]	0.224 [0.192, 0.259]	-0.028 [-0.056, -0.003]	0.287 [0.266, 0.318]
vit_b16	blur	0.073 [0.028, 0.112]	0.311 [0.266, 0.350]	0.000 [-0.021, 0.042]	0.378 [0.336, 0.402]
deit_s16	normalized-zero	0.091 [0.049, 0.119]	0.035 [-0.021, 0.087]	-0.157 [-0.189, -0.119]	0.378 [0.346, 0.406]
deit_s16	blur	0.101 [0.056, 0.133]	0.147 [0.112, 0.182]	-0.024 [-0.070, 0.014]	0.371 [0.332, 0.395]

Table S18.2. Type/depth-adjusted rank correlation and within-block head correlation. O, B, and O+B denote adjustment for operation, block, and both. Within each image, m and $|e|$ receive average ranks before least-squares residualization on an intercept and reference-coded one-hot fixed effects. Head correlations are computed within each block, clipped to $[-1+10^{-12}, 1-10^{-12}]$, averaged on the Fisher-z scale, and transformed back.

Model	Reference	A/V O	A/V B	A/V O+B	Residual O	Residual B	Residual O+B	Head within-block	A/V concordance
vit_b16	normalized-zero	0.122	-0.044	0.030	0.168	-0.010	-0.128	0.321	0.500
vit_b16	blur	0.164	-0.015	-0.018	0.184	0.134	0.042	0.303	0.500
deit_s16	normalized-zero	0.077	0.167	0.089	0.110	0.019	-0.091	0.245	0.583
deit_s16	blur	0.139	0.186	0.095	0.161	0.280	0.142	0.183	0.583

A/V concordance is the fraction of 12 blocks with positive $\mathrm{sign}[(m_A - m_V)(|e_A| - |e_V|)]$; zero products are excluded as ties. It is not Spearman and is reported descriptively without a sign test against zero. The v7 r4 root contains complete intervals, quantiles, per-block results, residual degrees of freedom, condition numbers, and independent validation. Every required correlation has 400/400 valid images.

The historical pooled association does not establish uniform within-type ranking. V-only is stronger in ViT-B, while the DeiT-S normalized-zero interval includes zero. MLP branches are positive in all four settings, whereas attention branches are negative or near zero. After joint type and block adjustment, A/V is near zero under both ViT-B references and residual correlations are negative under normalized zero; DeiT-S adjusted results change with reference.

Table S18.3. Own-token locality radius sensitivity; late-minus-early median [95% CI].

Model	Radius 0	Radius 1	Radius 2
vit_b16	-0.302 [-0.310, -0.298]	-0.356 [-0.363, -0.350]	-0.335 [-0.344, -0.322]
deit_s16	-0.260 [-0.263, -0.253]	-0.342 [-0.353, -0.333]	-0.315 [-0.323, -0.303]

All three radii retain the direction. Radius 1 remains the prespecified descriptive metric; this sensitivity analysis does not select a new primary metric.

The absolute-mass quartile diagnostic for residual output–identity TV uses images as statistical units. Median attention TV from Q1 to Q4 is 0.107→0.073 for ViT-B and 0.071→0.055 for DeiT-S; MLP values are 0.139→0.105 and 0.117→0.090. Spatial differences are not confined to the lowest-mass group, but this unadjusted post-hoc stratification does not exclude gradient quality or noise effects.

H4 was independently recomputed from 1,600 raw JSON/NPZ pairs. Medians for vit_b16, deit_s16, vit_b16_in1k, and vit_s16_augreg are -0.114, -0.186, -0.186, and -0.136. Positive/negative/tie counts are 101/297/2, 96/298/6, 56/342/2, and 107/285/8. All within-checkpoint Holm-adjusted values are below 10^{-6}. The v7 r4 root contains per-image and block-level correlations, full-precision p values, and machine receipts.

S19. Architecture cuts, entropy quality, and H4 aggregation sensitivity

S19.1 Three registries and certification of actual cuts

The machine contracts now separate scalar targets, local-differential decompositions, and finite-replacement interventions into three read-only derived registries. Row-level recounting of the original EP0 target registry gives 457/385 scalar targets per image for ViT-B/16 and DeiT-S/16, respectively. Each model has 192 post-block token-channel coordinates, 48 patch sums, 48 patch energies, 18 module patch aggregates, four patch-embedding aggregates, one final-CLS coordinate, and two class logits; head-coordinate counts are 144 and 72. Q/K/V-induced, routing/value, residual components, and intervenable variables are not included in these totals.

Registry	Contains	Explicitly excludes
scalar target	Scalar readout, graph location, aggregation rule	Path components and intervention variables
local-differential decomposition	Native use, analytic partition, cut, reconstruction object	Scalar-target counts and finite-replacement effects
finite-replacement intervention	Intervenable activation, reference, effect, and regional task	Local-differential closure objects

The frozen pre-LN ViT tensor-use DAG checks all 80,832 combinations of 842 targets, 12 blocks, and eight cut declarations. Dynamic programming over the target ancestor graph propagates the number of cut edges crossed by every input path. A full cut requires that count set to be exactly {1} and every declared edge to belong to the target ancestor graph. Of the pairs, 26,094 pass the full target-cut condition, 15,660 pass the attention branch-only partition

condition, 24,444 are rejected because the declared reconstruction object is inapplicable, and 14,634 because the target is upstream or the branch is absent from its ancestor graph. This proof covers the registry-bound standard unfused pre-LN ViT; numerical closure and mutation tests remain separate implementation evidence.

S19.2 Total-mass sensitivity of spatial entropy

Thresholds are defined by total-L1-mass distributions within each model and block over all frozen M3 unit maps. The original late-minus-early entropy is reproduced from source arrays with maximum error zero. The table reports image-level medians and 95% bootstrap intervals. Trimming retains all 100 images; quartile analyses propagate NA for a legitimately empty cell.

Model	Analysis	Median [95% CI]	Valid / total
ViT-B/16	no trimming	0.220 [0.211, 0.227]	100 / 100
ViT-B/16	trim lowest 1% mass	0.222 [0.213, 0.230]	100 / 100
ViT-B/16	trim lowest 5% mass	0.229 [0.221, 0.237]	100 / 100
ViT-B/16	trim lowest 10% mass	0.234 [0.227, 0.242]	100 / 100
DeiT-S/16	no trimming	0.225 [0.212, 0.228]	100 / 100
DeiT-S/16	trim lowest 1% mass	0.226 [0.215, 0.229]	100 / 100
DeiT-S/16	trim lowest 5% mass	0.230 [0.219, 0.235]	100 / 100
DeiT-S/16	trim lowest 10% mass	0.233 [0.223, 0.237]	100 / 100

Across model–block mass quartiles, ViT-B Q1–Q4 median differences are 0.090, 0.194, 0.238, and 0.266; DeiT-S values are 0.112, 0.205, 0.222, and 0.224. Eight image–quartile cells have an empty early or late group and are retained as typed NA, without deleting or replacing images. Median within-image, within-block Spearman association between log1p(total mass) and entropy is −0.729 [−0.747, −0.693] in ViT-B and −0.760 [−0.775, −0.723] in DeiT-S. Late-minus-early log1p mass is 2.893 [2.829, 2.961] and 2.183 [2.109, 2.227]. Entropy therefore depends on map quality, while the overall broadening direction is not produced solely by the lowest-mass maps.

S19.3 H4 matched aggregation and random-region precision

The matched path score is $M^{\mathrm{sum}}_{jbh} = |\sum_i \mathcal{L}_{jbh,i}|$, giving it the same sum-then-absolute aggregation order as the activation × gradient comparator. The target, normalized-zero reference, four blocks, finite-replacement effect, and image-level aggregation remain unchanged.

Checkpoint	Matched path minus activation × gradient	Positive / negative / tie
vit_b16	−0.124 [−0.147, −0.102]	89 / 311 / 0
deit_s16	−0.114 [−0.157, −0.079]	130 / 261 / 9
vit_b16_in1k	−0.226 [−0.253, −0.205]	49 / 349 / 2
vit_s16_augreg	−0.171 [−0.200, −0.143]	110 / 285 / 5

The original H4 is reproduced again from the same arrays with zero per-image maximum error. All four checkpoints remain negative under matched aggregation, so the absolute/sum ordering difference between scores does not explain the original result.

No additional model run is needed for the random-region baseline. Source validation covers 1,600 model–reference–image conditions and 32,000 repeat records, with all 20 repeats present in every condition. Median relative Monte

Carlo standard error is 5.1%–6.9%, with q90 of 7.6%–8.9%; 5/10/20-repeat resampling, split-half, and leave-one-repeat-out diagnostics are retained. This precision supports cohort-level comparisons; small single-image advantages are not decided from it.

References

Abnar, Samira, and Willem Zuidema. 2020. "Quantifying Attention Flow in Transformers." *ACL*.

Achtibat, Reduan, Sayed Mohammad Vakilzadeh Hatefi, Maximilian Dreyer, et al. 2024. "AttnLRP: Attention-Aware Layer-Wise Relevance Propagation for Transformers." *ICML*.

Buratti, Christopher, Michele Marchetti, Federico Parlapiano, Daniele Traini, Domenico Ursino, and Luca Virgili. 2026. *Gradient-Skipping Relevance Propagation for Efficient Explainability of Vision Transformers*. https://doi.org/10.48550/arXiv.2607.10365.

Chefer, Hila, Shir Gur, and Lior Wolf. 2021. "Transformer Interpretability Beyond Attention Visualization." *CVPR*.

Chen, Po-Kai, Aske Plaat, and Niki van Stein. 2026. *Every Component Is a Lookup: Token Attribution and Composition from a Single Decomposition*. https://doi.org/10.48550/arXiv.2605.23393.

Dhamdhere, Kedar, Mukund Sundararajan, and Qiqi Yan. 2018. "How Important Is a Neuron?" *arXiv Preprint arXiv:1805.12233*.

Dosovitskiy, Alexey, Lucas Beyer, Alexander Kolesnikov, et al. 2021. "An Image Is Worth 16x16 Words: Transformers for Image Recognition at Scale." *ICLR*.

Dou, Hongtao, Furao Shen, Jian Zhao, and Xiyu Mu. 2023. "Understanding Neural Network Through Neuron Level Visualization." *Neural Networks* 168: 484–95. https://doi.org/10.1016/j.neunet.2023.09.030.

Ferrando, Javier, Gerard I. Gallego, and Marta R. Costa-jussa. 2022. "Measuring the Mixing of Contextual Information in the Transformer." *EMNLP*.

Izadi, Amirhossein, Mohammad Banayeeanzade, Amin Mirrokni, et al. 2026. *Causal Attribution via Activation Patching*. https://doi.org/10.48550/arXiv.2603.13652.

Kramar, Janos, Tom Lieberum, Rohin Shah, and Neel Nanda. 2024. *AtP*: An Efficient and Scalable Method for Localizing LLM Behaviour to Components*. https://arxiv.org/abs/2403.00745.

Mehri, Faraz, Mahdieh Soleymani Baghshah, and Mohammad Taher Pilehvar. 2025. "LibraGrad: Balancing Gradient Flow for Universally Better Vision Transformer Attributions." *CVPR*.

Modarressi, Ali, Mohsen Fayyaz, Ehsan Aghazadeh, Yadollah Yaghoobzadeh, and Mohammad Taher Pilehvar. 2023. "DecompX: Explaining Transformers Decisions by Propagating Token Decomposition." *ACL*.

Modarressi, Ali, Mohsen Fayyaz, Yadollah Yaghoobzadeh, and Mohammad Taher Pilehvar. 2022. "GlobEnc: Quantifying Global Token Attribution by Incorporating the Whole Encoder Layer in Transformers." *NAACL-HLT*.

Nishankar, S., P. Sanjeewani, A. Perera, and S. Thuseethan. 2026. *HiLRP: Toward One Trustworthy Explanation for Vision Transformer: Conservation-Valid Attribution via Attention Primitives*. https://doi.org/10.48550/arXiv.2609.01282.

Selvaraju, Ramprasaath R., Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. 2017. "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization." *ICCV*.

Shrikumar, Avanti, Peyton Greenside, and Anshul Kundaje. 2017. "Learning Important Features Through Propagating Activation Differences." *ICML*.

Srinivas, Suraj, and Francois Fleuret. 2019. "Full-Gradient Representation for Neural Network Visualization." *NeurIPS*.

Sundararajan, Mukund, Ankur Taly, and Qiqi Yan. 2017. "Axiomatic Attribution for Deep Networks." *ICML*.

Syed, Aaquib, Can Rager, and Arthur Conmy. 2024. "Attribution Patching Outperforms Automated Circuit Discovery." *BlackboxNLP*. https://doi.org/10.18653/v1/2024.blackboxnlp-1.25.

Wrobel, Aleksandra, Siddhartha Gairola, Jacek Tabor, Bernt Schiele, Bartosz Zielinski, and Dawid Rymarczyk. 2026. *DAVE: Distribution-Aware Attribution via ViT Gradient Decomposition*. https://doi.org/10.48550/arXiv.2602.06613.

Zhang, Fred, and Neel Nanda. 2023. *Towards Best Practices of Activation Patching in Language Models: Metrics and Methods*. https://arxiv.org/abs/2309.16042.

Zukowska, Natalia, Wolfgang Stammer, Bernt Schiele, and Jonas Fischer. 2026. *Seeing Through Circuits: Faithful Mechanistic Interpretability for Vision Transformers*. https://doi.org/10.48550/arXiv.2604.14477.